# Non-commutative algebra

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Abstract. The notes correspond to the master course **Non-commutative algebra** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences.

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## Introduction

The notes correspond to the master course **Non-commutative Algebra** of the Vrije Universiteit Brussel, Faculty of Sciences, Department of Mathematics and Data Sciences. The course is divided into twelve two-hour lectures.

Most of the material is based on standard results from group theory, primarily finite group theory. The main references are [9], [12], and [13]. However, the course also covers topics not included in these texts, and additional references are provided where appropriate.

The notes include many exercises, some with full detailed solutions. Mandatory exercises have a green background, while optional ones (bonus exercises) have a yellow background.

The notes also include some additional comments. While these are entirely optional, I hope they offer further insight. They are highlighted with a pink background.

The notes include Magma code, which we use to verify examples and offer alternative solutions to certain exercises. Magma [2] is a powerful software tool designed for working with algebraic structures. There is a free online version of Magma available.

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## 1. Solvable groups

A subgroup $H$ of $G$ is said to be **characteristic** if $f(H) \subseteq H$ for all $f \in \mathrm{Aut}(G)$. The center and the commutator subgroup are characteristic subgroups. Every characteristic subgroup is normal, as the maps $x \mapsto gxg^{-1}$ are automorphisms.

1.1. Exercise. Prove that if $H$ is characteristic in $K$ and $K$ is normal in $G$, then $H$ is normal in $G$.

For a group $G$ and $x, y, z \in G$, conjugation will be considered as a left action of $G$ on $G$ and we will use the following notation: ${}^{x}y = xyx^{-1}$. The commutator between $x$ and $y$ will be written as

$$[x, y] = xyx^{-1}y^{-1} = ({}^{x}y)y^{-1}.$$

We will also use the following notation:

$$[x, y, z] = [x, [y, z]].$$

For subgroups $H$ and $K$ of $G$, let

$$[H, K] = \langle [h, k] : h \in H,\ k \in K \rangle.$$

Note that $[X, Y] = [Y, X]$. For subgroups $X$, $Y$ and $Z$ of $G$, we write

$$[X, Y, Z] = [X, [Y, Z]]\,.$$

For a group $G$, let $G^{(0)} = G$ and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$ for $i \geq 0$. The **derived series** of $G$ is the sequence

$$G = G^{(0)} \supseteq G^{(1)} \supseteq G^{(2)} \supseteq \cdots$$

Each $G^{(i)}$ is a characteristic subgroup of $G$.

1.2. Definition. We say that a group $G$ is **solvable** if $G^{(n)} = \{1\}$ for some $n$.

1.3. Example. Abelian groups are solvable.

1.4. Example. The group $\mathbf{SL}_2(3)$ is solvable, as the derived series is

$$\mathbf{SL}_2(3) \supseteq Q_8 \supseteq C_4 \supseteq \{1\}.$$

1.5. Example. Non-abelian simple groups cannot be solvable.

1.6. Exercise. Let $G$ be a group. Prove the following statements:

**1)** A subgroup $H$ of $G$ is solvable, when $G$ is solvable.

**2)** Let $K$ be a normal subgroup of $G$. Then $G$ is solvable if and only if $K$ and $G/K$ are solvable.

1.7. Example. For $n \geq 5$ the group $\mathbb{A}_n$ is simple and non-abelian. Hence it is not solvable. It follows that $\mathbb{S}_n$ is not solvable for $n \geq 5$.

1.8. Exercise. Let $p$ be a prime number. Prove that finite $p$-groups are solvable.

1.9. Exercise. Let $p$, $q$ and $r$ be prime numbers. Prove that groups of order $p^\alpha q$, $p^2q^2$ and $pqr$ are solvable.

1.10. Exercise. Prove that groups of order $< 60$ are solvable.

1.11. Theorem (Burnside). *Let $p$ be a prime number. If $G$ is a finite group that has a conjugacy class with $p^k > 1$ elements, then $G$ is not simple.*

The easiest way to prove Theorem 1.11 is using character theory.

1.12. Theorem (Burnside). *Let $p$ and $q$ be prime numbers. If $G$ has order $p^a q^b$, then $G$ is solvable.*

1.13. Theorem (Feit–Thompson). *Groups of odd order are solvable.*

1.14. Definition. Let $p$ be a prime number. A $p$-group $P$ is said to be **elementary abelian** if $x^p = 1$ for all $x \in P$.

1.15. Definition. A subgroup $M$ of $G$ is said to be **minimal normal** if $M \neq \{1\}$ (or $G = \{1\}$), $M$ is normal in $G$ and the only normal subgroup of $G$ properly contained in $M$ is $\{1\}$.

1.16. Example. If a normal subgroup $M$ is minimal (with respect to the inclusion), then it is minimal and normal. However, the converse statement does not hold. For example, the subgroup of $\mathbb{A}_4$ generated by (12)(34), (13)(24) and (14)(23) is minimal normal in $\mathbb{A}_4$ but it is not minimal.

1.17. Exercise. Prove that every finite group contains a minimal normal subgroup.

1.18. Example. Let $G = \mathbb{D}_6 = \langle r, s : r^6 = s^2 = 1, srs = r^{-1} \rangle$ the dihedral group of twelve elements. The subgroups $S = \langle r^2 \rangle$ and $T = \langle r^3 \rangle$ are (the only) minimal normal in $G$.

1.19. Example. Let $G = \mathbf{SL}_2(3)$. The only minimal normal subgroup of $G$ is its center $Z(\mathbf{SL}_2(3)) \simeq C_2$.

The following lemma will be very useful later.

1.20. Lemma. *Let $M$ be a minimal normal subgroup of $G$. If $M$ is solvable and finite, then $M$ is an elementary abelian $p$-group for some prime number $p$.*

Proof. Since $M$ is solvable, $[M, M] \subsetneq M$. Moreover, $[M, M]$ is normal in $G$, as $[M, M]$ is characterisitic in $M$ and $M$ is normal in $G$. Since $M$ is minimal normal, $[M, M] = \{1\}$. Hence $M$ is abelian.

Since $M$ is finite, there is a prime number $p$ such that $\{1\} \neq P = \{x \in M : x^p = 1\} \subseteq M$. Since $P$ is characteristic in $M$, $P$ is normal in $G$. By minimality, $P = M$. □

1.21. Theorem. *Let $G$ be a finite non-trivial solvable group. Then every maximal subgroup of $G$ has index $p^\alpha$ for some prime number $p$.*

Proof. We proceed by induction on $|G|$. If $|G|$ is a prime power, the claim is clear. Assume that $|G| \geq 6$ and let $M$ be a maximal subgroup of $G$. Let $N$ be a minimal normal subgroup of $G$ and $\pi \colon G \to G/N$ the canonical map. If $N = G$, then $N = G$ is a $p$-group and we are done. Assume then that $N \neq G$. Since $M \subseteq NM \subseteq G$, either $M = NM$ or $NM = G$ (by the maximality of $M$). If $M = NM \supseteq N$, then $\pi(M)$ is a maximal subgroup of $\pi(G) = G/N$. Hence

$$(G : M) = (\pi(G) : \pi(M))$$

is a prime power by the inductive hypothesis. If $NM = G$, then

$$(G : M) = \frac{|G|}{|M|} = \frac{|NM|}{|M|} = \frac{|N|}{|N \cap M|}$$

is a prime power, because $N$ is a $p$-group by the previous lemma. □

1.22. Exercise. Let $G$ be a finite non-trivial solvable group. Prove that there exists a prime number $p$ such that $G$ contains a minimal normal $p$-subgroup.

1.23. Example. Let $G = \mathbb{S}_4$. The 2-subgroup

$$K = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\} \simeq C_2 \times C_2$$

is minimal normal. Note that $G$ does not have minimal normal 3-subgroups.

1.24. Theorem. *Let $G$ be a finite non-trivial group. Then $G$ is solvable if and only if every non-trivial quotient of $G$ contains an abelian non-trivial normal subgroup.*

Proof. Every quotient of $G$ is solvable and therefore contains an abelian minimal normal subgroup. To prove the converse we proceed by induction on $|G|$. Let $N$ be a normal abelian subgroup of $G$. If $N = G$, then $G$ is solvable (because it is abelian). If $N \neq G$, then $|G/N| < |G|$. Since every quotient of $G/N$ is a quotient of $G$, the group $G/N$ satisfies the assumptions of the theorem. Hence $G/N$ is solvable by the inductive hypothesis. Now $N$ and $G/N$ are solvable, so is $G$. □

1.25. Exercise. Let $G$ be a group. Prove that $G$ is solvable if and only if there is a sequence

$$\{1\} = N_0 \subseteq N_1 \subseteq \cdots \subseteq N_k = G$$

of normal subgroups such that every quotient $N_i/N_{i-1}$ is abelian.

## 2. Hall's theorem

We start with an extremely simple and useful tool.

2.1. Lemma (Frattini's argument). *Let $G$ be a finite group and $K$ be a normal subgroup of $G$. If $P \in \mathrm{Syl}_p(K)$ for some prime number $p$, then $G = KN_G(P)$.*

Proof. Let $g \in G$. Since $gPg^{-1} \subseteq gKg^{-1} = K$, because $K$ is normal in $G$, and $gPg^{-1} \in \mathrm{Syl}_p(K)$, there exists $k \in K$ such that $kPk^{-1} = gPg^{-1}$. Hence $k^{-1}g \in N_G(P)$, as $P = (k^{-1}g)P(k^{-1}g)^{-1}$. Therefore $g = k(k^{-1}g) \in KN_G(P)$. □

2.2. Theorem (Hall). *Let $G$ be a finite group such that every maximal subgroup of $G$ has a prime or a prime-square index. Then $G$ is solvable.*

Proof. We proceed by induction on $|G|$. Let $p$ be the largest prime divisor of $|G|$. Let $S \in \mathrm{Syl}_p(G)$ and $N = N_G(S)$.

If $N = G$, then $S$ is normal in $G$. Since every maximal subgroup of $G/S$ has prime or a prime-square index, $G/S$ is solvable by the inductive hypothesis. Since $S$ is a $p$-group, it is solvable. Therefore $G$ is solvable.

Assume now that $N \neq G$. Let $H$ be a maximal subgroup of $G$ containing $N$. Then

$$N = N_H(S) = N_G(S),$$

the number of Sylow $p$-subgroups of $G$ is $(G : N) \equiv 1 \bmod p$ and the number of Sylow $p$-subgroups of $H$ is $(H : N) \equiv 1 \bmod p$ by the third Sylow's theorem. So,

$$\underbrace{(G : N)}_{\equiv 1 \bmod p} = (G : H)\,\underbrace{(H : N)}_{\equiv 1 \bmod p},$$

implies that also $(G : H) \equiv 1 \bmod p$. By assumption, there exists a prime number $q$ such that $(G : H) \in \{q, q^2\}$. Since $q$ divides $|G|$, it follows that $q < p$. If $(G : H) = q$, then $p$ divides $q - 1$ and therefore $p \leq q - 1 < p$, a contradiction. Thus $q^2 = (G : H) \equiv 1 \bmod p$. From this it follows that $q \equiv -1 \bmod p$ and hence $q = 2$ and $p = q + 1 = 3$.

Therefore $G$ has order $2^\alpha 3^\beta$. If we apply Burnside's theorem, we are done. Instead, we will finish the proof with an elementary argument. Let $K$ be a minimal normal subgroup of $G$. By Frattini's argument (Lemma 2.1), $G = KN = KH$. Since $H$ is maximal,

$$(K : K \cap H) = (G : H) = 4,$$

as $(G : H) = |G|/|H| = |KH|/|H| = |K|/|K \cap H|$. Since $(K : K \cap H) = 4$, letting $K$ act on $K/K \cap H$ by left multiplication, there exists a non-trivial group homomorphism $\rho\colon K \to \mathbb{S}_4$. Since $[K, K]$ is characteristic in $K$ and $K$ is normal in $G$, $[K, K] \subseteq K$ is normal in $G$. Since $K$ is minimal normal in $G$, there are two possible cases: either $[K, K] = \{1\}$ or $[K, K] = K$. If $[K, K] = K$, since $\mathbb{S}_4$ is solvable, $\rho(K)$ is solvable. Then

$$\rho(K) = \rho([K, K]) = [\rho(K), \rho(K)],$$

a contradiction. Therefore $[K, K] = \{1\}$ and $K$ is solvable (as it is abelian). □

## 3. Wielandt's solvability theorem

3.1. Lemma. *Let $G$ be a finite group and $H$ and $K$ be subgroups of $G$ of coprime indices. Then $G = HK$ and $(H : H \cap K) = (G : K)$.*

Proof. Let $D = H \cap K$. Since

$$(G : D) = \frac{|G|}{|H \cap K|} = (G : H)(H : H \cap K),$$

$(G : H)$ divides $(G : D)$. Similarly, $(G : K)$ divides $(G : D)$. Since $(G : H)$ and $(G : K)$ are coprime, $(G : H)(G : K)$ divides $(G : D)$. In particular,

$$\frac{|G|}{|H|}\frac{|G|}{|K|} = (G : H)(G : K) \leq (G : D) = \frac{|G|}{|H \cap K|}$$

and hence $|G| = |HK|$. Since

$$|G| = |HK| = |H||K|/|H \cap K|,$$

we conclude that $(G : K) = (H : H \cap K)$. □

3.2. Definition. Let $G$ be a group and $H$ be a subgroup of $G$. The **normal closure** $H^G$ of $H$ in $G$ is the subgroup $H^G = \langle xHx^{-1} : x \in G \rangle$.

3.3. Exercise. Let $G$ be a group and $H$ a subgroup of $G$. Prove that $H^G$ is normal in $G$ and that $H^G$ is the smallest normal subgroup of $G$ containing $H$.

3.4. Example. Let $G = \mathbb{A}_4$ and $H = \{\mathrm{id}, (12)(34)\}$. The normal closure of $H$ in $G$ is

$$H^G = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\} \simeq C_2 \times C_2.$$

3.5. Theorem (Wielandt's solvability theorem). *Let $G$ be a finite group and $H$, $K$ and $L$ be subgroups of $G$ with pairwise coprime indices. If $H$, $K$ and $L$ are solvable, then $G$ is solvable.*

Proof. Let $G$ be a minimal counterexample. Then $G \neq \{1\}$. There are two cases to consider. Assume first that $H = \{1\}$. Then $|G| = (G : H)$ is coprime with $(G : K)$ and thus $G = K$ is solvable. Assume now that $H \neq \{1\}$. Let $M$ be a minimal normal subgroup of $H$. By Lemma 1.20, $M$ is a $p$-group for some prime number $p$. Without loss of generality, we may assume that $p$ does not divide $(G : K)$ (otherwise, if $p$ divides $(G : K)$, then $p$ does not divide $(G : L)$ and we just need to change $K$ by $L$). There exists $P \in \mathrm{Syl}_p(G)$ such that $P \subseteq K$. Sylow subgroups are conjugate, so there exists $g \in G$ such that $M \subseteq gKg^{-1}$. Since $(G : gKg^{-1}) = (G : K)$ is coprime with $(G : H)$, Lemma 3.1 implies that $G = (gKg^{-1})H$.

We claim that all conjugates of $M$ are in $gKg^{-1}$. If $x \in G$, write $x = uv$ some some $u \in gKg^{-1}$ and $v \in H$. Since $M$ is normal in $H$,

$$xMx^{-1} = (uv)M(uv)^{-1} = uMu^{-1} \subseteq gKg^{-1}.$$

Let $N = M^G$ be the normal closure of $M$ in $G$. Then $\{1\} \neq N \subseteq gKg^{-1}$ is solvable, as $gKg^{-1}$ is solvable. We claim that $G/N$ is solvable. Let $\pi \colon G \to G/(M^G)$ be the canonical map. Since $H$, $K$ and $L$ are solvable, the subgroups $\pi(H)$, $\pi(K)$ and $\pi(L)$ of $\pi(G)$ are solvable. By the correspondence theorem, $\pi(H)$, $\pi(K)$ and $\pi(L)$ have pairwise coprime indices. Moreover, $\pi(G)$ is solvable, as $|\pi(G)| < |G|$. Hence $G$ is solvable. □

## 4. Hall's theorem

4.1. Definition. Let $G$ be a finite group of order $p^\alpha m$, where $p$ is a prime number such that $\gcd(p, m) = 1$. A subgroup $H$ of $G$ is said to be a **$p$-complement** if $|H| = m$.

4.2. Example. Let $G = \mathbb{S}_3$. Then $H = \langle (123) \rangle$ is a 2-complement and $K = \langle (12) \rangle$ is a 3-complement.

4.3. Theorem (Hall). *Let $G$ be a finite group that admits a $p$-complement for every prime divisor $p$ of $|G|$. Then $G$ is solvable.*

Proof. Let $|G| = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$ with $p_1 < \cdots < p_k$ prime numbers. We proceed by induction on $k$. If $k = 1$, then the claim holds, as $G$ is a $p$-group. If $k = 2$, the result holds by Burnside's theorem. Assume now that $k \geq 3$. For $j \in \{1, 2, 3\}$, let $H_j$ be a $p_j$-complement in $G$. Since $|H_j| = |G|/p_j^{\alpha_j}$, the subgroups $H_j$ have pairwise coprime indices.

We claim that $H_1$ is solvable. Note that $|H_1| = p_2^{\alpha_2} \cdots p_k^{\alpha_k}$. Let $p$ be a prime number dividing $|H_1|$ and $Q$ be a $p$-complement in $G$. Since $(G : H_1)$ and $(G : Q)$ are coprime, Lemma 3.1 implies that

$$(H_1 : H_1 \cap Q) = (G : Q).$$

Then $H_1 \cap Q$ is a $p$-complement in $H_1$. Therefore $H_1$ is solvable by the inductive hypothesis. Similarly, both $H_2$ and $H_3$ are solvable.

Since $H_1$, $H_2$ and $H_3$ are solvable of pairwise coprime indices, the theorem follows from Wielandt's theorem. □

## 5. Nilpotent groups

5.1. Exercise (The Hall–Witt identity). Let $G$ be a group and $x, y, z \in G$. Prove that

$$\left({}^y[x, y^{-1}, z]\right) \left({}^z[y, z^{-1}, x]\right) \left({}^x[z, x^{-1}, y]\right) = 1. \tag{5.1}$$

If $G$ is a group and $[G, G]$ is central in $G$, then the Hall-Witt identity becomes Jacobi's identity.

5.2. Lemma (Hall's three subgroups lemma). *Let $X$, $Y$ and $Z$ be subgroups of $G$ such that $[X, Y, Z] = [Y, Z, X] = \{1\}$. Then $[Z, X, Y] = \{1\}$.*

Proof. Since $[X, Y] \subseteq C_G(Z)$, it is enough to prove that $[z, x^{-1}, y] = 1$ for all $x \in X$, $y \in Y$ and $z \in Z$. Since $[y^{-1}, z] \in [Y, Z]$, $[x, y^{-1}, z] \in [X, Y, Z] = \{1\}$. Thus ${}^y[x, y^{-1}, z] = 1$. Similarly, ${}^z[y, z^{-1}, x] = 1$. Using the Hall–Witt identity, we conclude that $[z, x^{-1}, y] = 1$. □

5.3. Exercise. Let $N$ be a normal subgroup of $G$ and $X$, $Y$ and $Z$ be subgroups of $G$. If $[X, Y, Z] \subseteq N$ and $[Y, Z, X] \subseteq N$, then $[Z, X, Y] \subseteq N$.

5.4. Definition. Let $G$ be a group. The **lower central series** is the sequence $\gamma_k(G)$ of subgroups defined inductively as

$$\gamma_1(G) = G, \quad \gamma_{i+1}(G) = [G, \gamma_i(G)] \quad i \geq 1.$$

5.5. Definition. A group $G$ is said to be **nilpotent** if there exists a positive integer $c$ such that $\gamma_{c+1}(G) = \{1\}$. The smallest $c$ with $\gamma_{c+1}(G) = \{1\}$ is the **nilpotency class** of $G$.

5.6. Exercise. Prove that every nilpotent group is solvable.

A group is nilpotent of nilpotency class one if and only if it is abelian.

5.7. Example. The group $\mathbb{S}_3$ is solvable, as $\mathbb{S}_3 \supseteq \mathbb{A}_3 \supseteq \{1\}$ is a composition series with abelian factors. However, $\mathbb{S}_3$ is not nilpotent, as

$$\gamma_1(\mathbb{S}_3) = \mathbb{A}_3, \quad \gamma_2(\mathbb{S}_3) = [\mathbb{A}_3, \mathbb{S}_3] = \mathbb{A}_3,$$

and therefore $\gamma_i(\mathbb{S}_3) \neq \{1\}$ for all $i \geq 1$.

5.8. Example. The group $G = \mathbb{A}_4$ is not nilpotent, as

$$\gamma_1(G) = G, \quad \gamma_j(G) = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\} \simeq C_2 \times C_2$$

for all $j \geq 2$. We can do this with the computer:

```
gap> IsNilpotent(AlternatingGroup(4));
false
```

Let us do the calculation of the lower central series with the computer:

```
gap> List(LowerCentralSeries(AlternatingGroup(4)),\
StructureDescription);
[ "A4", "C2 x C2" ]
```

Here is an alternative:

```
gap> G := AlternatingGroup(4);;
gap> gamma_1 := G;;
gap> gamma_2 := DerivedSubgroup(G);;
gap> gamma_3 := CommutatorSubgroup(gamma_2,G);;
gap> StructureDescription(gamma_1);
"A4"
gap> StructureDescription(gamma_2);
"C2 x C2"
gap> StructureDescription(gamma_3);
"C2 x C2"
```

5.9. Exercise. Let $G$ be a group. Prove the following statements:

**1)** Each $\gamma_i(G)$ is a characteristic subgroup of $G$.
**2)** $\gamma_i(G) \supseteq \gamma_{i+1}(G)$ for all $i \geq 1$.
**3)** If $f\colon G \to H$ is a surjective group homomorphism, then $f(\gamma_i(G)) = \gamma_i(H)$ for all $i \geq 1$.

5.10. Exercise. Prove that if $H$ and $K$ are nilpotent groups, then $H \times K$ is nilpotent.

5.11. Exercise. Let $G$ be a nilpotent group. Prove the following statements:

**1)** Subgroups of $G$ are nilpotent.
**2)** If $f\colon G \to H$ is a surjective homomorphism, then $H$ is nilpotent.

5.12. Exercise. True or false? If $G$ is a nilpotent group and $N$ is normal subgroup of $G$ such that $N$ and $G/N$ are nilpotent, then $G$ is nilpotent.

5.13. Proposition. *Finite $p$-groups are nilpotent.*

Proof. We proceed by induction on $|G|$. The case $G = \{1\}$ is trivial. Assume the result holds for $p$-groups of order $< |G|$. Since $G$ is a $p$-group, $Z(G) \neq \{1\}$. By the inductive hypothesis, $G/Z(G)$ is nilpotent. There exists $c$ such that $\gamma_{c+1}(G/Z(G)) = \{1\}$.

Let $\pi\colon G \to G/Z(G)$ be the canonical map. By Exercise 5.9,

$$\pi(\gamma_{c+1}(G)) = \gamma_{c+1}(G/Z(G)) = \{1\}.$$

Then $\gamma_{c+1}(G) \subseteq \ker \pi = Z(G)$. Hence $G$ is nilpotent, as

$$\gamma_{c+2}(G) = [G, \gamma_{c+1}(G)] = [G, Z(G)] = \{1\}. \qquad \square$$

5.14. Theorem. *If $G$ is a group, then $[\gamma_i(G), \gamma_j(G)] \subseteq \gamma_{i+j}(G)$ for all $i, j \geq 1$.*

Proof. We proceed by induction on $i$. The case $i = 1$ is trivial, as $[G, \gamma_j(G)] = \gamma_{j+1}(G)$ by definition. Assume that the result holds for some $i \geq 1$ and all $j \geq 1$.

First note that

$$[G, \gamma_i(G), \gamma_j(G)] \subseteq [G, \gamma_{i+j}(G)] = \gamma_{i+j+1}(G)$$

by the inductive hypothesis. Moreover, by the inductive hypothesis,

$$[\gamma_i(G), \gamma_j(G), G] = [\gamma_i(G), G, \gamma_j(G)] = [\gamma_i(G), \gamma_{j+1}(G)] \subseteq \gamma_{i+j+1}(G)$$

By using Exercise 5.3 with $X = G$, $Y = \gamma_i(G)$ and $Z = \gamma_j(G)$, we get that

$$[\gamma_j(G), G, \gamma_i(G)] \subseteq \gamma_{i+j+1}(G).$$

Hence

$$[\gamma_{i+1}(G), \gamma_j(G)] = [\gamma_j(G), \gamma_{i+1}(G)] = [\gamma_j(G), G, \gamma_i(G)] \subseteq \gamma_{i+j+1}(G). \qquad \square$$

We may have to consider commutators with $n$ copies of the group that are not necessarily associated on the right. For example, if $n = 3$, we might have something like $[G, G, G] = [G, [G, G]]$ or $[[G, G], G]$. These expressions will be called commutators of **weight** three.

5.15. Corollary. *In a group $G$, every weight $n$ commutator is contained in $\gamma_n(G)$.*

Proof. We proceed by induction on $n$. The case $n = 1$ is trivial. Assume that $n \geq 1$ and the result holds for all $j \leq n$. An arbitrary commutator of weight $n + 1$ is of the form $[A, B]$, where $A$ is a commutator of weight $k$, $B$ is a commutator of weight $l$ and $n+1 = k+l$. Since $k \leq n$ and $l \leq n$, the inductive hypothesis implies that $A \subseteq \gamma_k(G)$ and $B \subseteq \gamma_l(G)$. Hence $[A, B] \subseteq [\gamma_k(G), \gamma_l(G)] \subseteq \gamma_{k+l}(G)$ by the previous theorem. $\square$

5.16. Exercise. Let $G$ be a group. Prove that $G^{(k)} \subseteq \gamma_{2^k}(G)$ for all $k \geq 1$.

5.17. Exercise. Let $G$ be a nilpotent group of class $m$. Prove that the derived length (i.e. the length of the derived series) of $G$ is $\leq 1 + \log_2 m$.

The following lemma is important. It states that nilpotent groups satisfy the **normalizer condition**.

5.18. Lemma (normalizer condition). *Let $G$ be a nilpotent group. If $H$ is a proper subgroup of $G$, then $H \subsetneq N_G(H)$.*

Proof. There exists $c$ such that $G = \gamma_1(G) \supseteq \cdots \supseteq \gamma_{c+1}(G) = \{1\}$. Since $\{1\} = \gamma_{c+1}(G) \subseteq H$ and $\gamma_1(G) \not\subseteq H$, let $k$ be the smallest positive integer such that $\gamma_k(G) \subseteq H$. Since

$$[H, \gamma_{k-1}(G)] \subseteq [G, \gamma_{k-1}(G)] = \gamma_k(G) \subseteq H,$$

we obtain that $xHx^{-1} \subseteq H$ for all $x \in \gamma_{k-1}(G)$, that is $\gamma_{k-1}(G) \subseteq N_G(H)$. If $N_G(H) = H$, then $\gamma_{k-1}(G) \subseteq H$, a contradiction to the minimality of $k$. □

For a group $G$, we define the sequence $\zeta_0(G), \zeta_1(G), \ldots$ recursively as follows:

$$\zeta_0(G) = \{1\}, \quad \zeta_{i+1}(G) = \{g \in G : [x, g] \in \zeta_i(G) \text{ for all } x \in G\}, \quad i \geq 0.$$

For example, $\zeta_1(G) = Z(G)$.

5.19. Lemma. *Let $G$ be a group. For every $i \geq 0$, the set $\zeta_i(G)$ is a normal subgroup of $G$.*

Proof. We proceed by induction on $i$. The case $i = 0$ is trivial, as $\zeta_0(G) = \{1\}$. Assume the result holds for some $i$. We claim that $\zeta_{i+1}(G)$ is normal subgroup of $G$. Let $g, h \in \zeta_{i+1}(G)$ and $x \in G$. By the inductive hypothesis,

$$[x, g^{-1}] = (xg^{-1})[x^{-1}, g](xg^{-1})^{-1} \in \zeta_i(G)(xg^{-1})^{-1} = \zeta_i(G),$$
$$[x, gh] = [x, h][hxh^{-1}, g] \in \zeta_i(G).$$

Since $1 \in \zeta_{i+1}(G)$, we conclude that each $\zeta_i(G)$ is a subgroup of $G$. Moreover, since

$$[xgx^{-1}, y] = x[g, x^{-1}yx]x^{-1} \in \zeta_i(G)$$

for all $y \in G$, we obtain that $xgx^{-1} \in \zeta_{i+1}(G)$. □

5.20. Definition. Let $G$ be a group. The **ascending central series** of $G$ is the sequence

$$\{1\} = \zeta_0(G) \subseteq \zeta_1(G) \subseteq \zeta_2(G) \subseteq \cdots$$

5.21. Definition. A group $G$ is said to be **perfect** if $[G, G] = G$.

5.22. Theorem (Grün). *If $G$ is a perfect group, then $Z(G/Z(G)) = \{1\}$.*

Proof. By definition, $[G, \zeta_2(G)] \subseteq Z(G)$ and $[\zeta_2(G), G] \subseteq Z(G)$. Then

$$[G, G, \zeta_2(G)] = [G, \zeta_2(G), G] = \{1\}.$$

By using the three subgroups lemma with $X = Y = G$ and $Z = \zeta_2(G)$,

$$[\zeta_2(G), G] = [\zeta_2(G), [G, G]] = [\zeta_2(G), G, G] = \{1\}.$$

Thus $\zeta_2(G) \subseteq Z(G)$.

We aim to prove that $Z(G/Z(G))$ is trivial. Let $\pi\colon G \to G/Z(G)$ be the canonical map and $g \in G$ be such that $\pi(g)$ is central. Since

$$1 = [\pi(x), \pi(g)] = \pi([x, g])$$

for all $x \in G$, $[x, g] \in Z(G) = \zeta_1(G)$ for all $x \in G$. Hence $g \in \zeta_2(G) \subseteq Z(G)$. □

## 6. Berkovich's theorem

We start with an exercise on the **normal core** of a subgroup.

6.1. Exercise. Let $G$ be a group and $A$ be a subgroup of $G$. Prove that

$$\mathrm{Core}_G(A) = \bigcap_{x\in G} xAx^{-1}$$

is the largest normal subgroup of $G$ contained in $A$.

6.2. Theorem (Berkovich). *Let $G$ be a finite solvable group and $H$ be a proper subgroup of $G$ with the smallest possible index. Then $H$ is normal in $G$.*

Proof. Note that $H$ is a maximal subgroup. Let $C = \mathrm{Core}_G(H)$. There are two cases to consider.

Assume first that $C \neq \{1\}$. Let $\pi\colon G \to G/C$ be the canonical map. By the inductive hypothesis, $\pi(H)$ is normal in $\pi(G) = G/C$, as by the correspondence theorem,

$$(\pi(G) : \pi(H)) = (G : H)$$

is the smallest possible index in $G/C$. Thus $H$ is normal in $G$ again by the correspondence theorem.

Assume now that $C = \{1\}$. Let $N$ be a minimal normal subgroup of $G$. Since $N$ is abelian, $N \cap H$ is normal in $N$. Moreover, since $N$ is normal in $G$, $N \cap H$ is normal in $H$. Thus $N \cap H$ is a normal subgroup of $NH$.

We claim that $N \not\subseteq H$. If $N \subseteq H$, then, for every $x \in G$,

$$N = xNx^{-1} \subseteq xHx^{-1}.$$

Therefore

$$N \subseteq \bigcap_{x\in G} xHx^{-1} = \mathrm{Core}_G(H) = \{1\},$$

a contradiction.

Since $H$ is maximal, either $NH = G$ or $NH = H$. If $NH = H$, then $N \subseteq NH = H$, a contradiction. Hence $NH = G$. Thus $N \cap H$ is a normal subgroup of $NH = G$. The minimality of $N$ implies then that $N \cap H = \{1\}$ and therefore

$$|N||H| = \frac{|N||H|}{|N \cap H|} = |G|.$$

This implies that $|N| = (G : H)$.

Let $H$ act on $N$ by conjugation. By the fundamental counting principle,

$$|H \cdot x| = (H : H_x)$$

for all $x \in N$. Since $H \cdot 1 = \{1\}$, every orbit has at most $|N| - 1$ elements, that is

$$(H : H_x) = |H \cdot x| \leq |N| - 1.$$

for all $x \in N$.

Since $N$ is a normal subgroup of $G$, the set $NH_x$ is a subgroup of $G$. Moreover,

$$N \cap H_x \subseteq N \cap H = \{1\}.$$

Thus

$$(G : NH_x) = \frac{|G|}{|NH_x|} = \frac{|G|}{|N||H_x|} = \frac{|N||H|}{|N||H_x|} = (H : H_x) \leq |N| - 1 = (G : H) - 1,$$

a contradiction. □

The solution of the following exercise is implicit in the proof of Berkovich's theorem.

6.3. Exercise. Let $G$ be a finite solvable group and $H$ a subgroup of $G$. Assume that $H$ is contained in maximal subgroup $M$ such that $\mathrm{Core}_G(M) = \{1\}$. Prove that $G$ has a subgroup with index equal to $(M : H)$.

Let $G$ be a group and $K$ be a subgroup of $G$. We say that $K$ **normalizes** $H$ if $K \subseteq N_G(H)$. We say that $K$ **centralizes** $H$ if $K \subseteq C_G(H)$, that is if and only if $[H, K] = \{1\}$.

6.4. Exercise. Let $K$ and $H$ be subgroups of $G$ such that $K \subseteq H$ and $K$ is normal in $G$. Prove that $[H, G] \subseteq K$ if and only if $H/K \subseteq Z(G/K)$.

6.5. Lemma. *Let $G$ be a group. There exists an integer $c$ such that $\zeta_c(G) = G$ if and only if $\gamma_{c+1}(G) = \{1\}$. In this case,*

$$\gamma_{i+1}(G) \subseteq \zeta_{c-i}(G)$$

*for all $i \in \{0, 1, \ldots, c\}$.*

Proof. Assume first that $\zeta_c(G) = G$. To prove that $\gamma_{i+1}(G) \subseteq \zeta_{c-i}(G)$ holds for all $i$, we proceed by induction. The case $i = 0$ is trivial. So assume that the result holds for some $i \geq 0$. If $g \in \gamma_{i+2}(G) = [G, \gamma_{i+1}(G)]$, then

$$g = \prod_{k=1}^{N} [x_k, g_k],$$

for some $g_1, \ldots, g_N \in \gamma_{i+1}(G)$ and $x_1, \ldots, x_N \in G$. By the inductive hypothesis,

$$g_k \in \gamma_{i+1}(G) \subseteq \zeta_{c-i}(G)$$

for all $k$. Hence $[x_k, g_k] \in \zeta_{c-i-1}(G)$ for all $k$. Therefore $g \in \zeta_{c-(i+1)}(G)$.

We now assume that $\gamma_{c+1}(G) = \{1\}$. We aim to prove that $\gamma_{i+1}(G) \subseteq \zeta_{c-i}(G)$ holds for all $i$. We proceed by backward induction on $i$. The case $i = c$ is trivial. So assume the result holds for some $i + 1 \leq c$. Let $g \in \gamma_i(G)$. By the inductive hypothesis,

$$[x, g] \in [G, \gamma_i(G)] = \gamma_{i+1}(G) \subseteq \zeta_{c-i}(G).$$

Thus $g \in \zeta_{c-i+1}(G)$ by definition. □

6.6. Example. Let $G = \mathbb{S}_3$. Then $\zeta_j(G) = \{1\}$ for all $j \geq 0$:

```
gap> UpperCentralSeries(SymmetricGroup(3));
[ Group(()) ]
```

6.7. Definition. Let $G$ be a group. A **central series** for $G$ is a sequence

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$$

of normal subgroups of $G$ such that for each $i \in \{1, \ldots, n\}$, $\pi_i(G_{i-1})$ is a subgroup of $Z(G/G_i)$, where $\pi_i \colon G \to G/G_i$ is the canonical map.

6.8. Lemma. *Let $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$ be a central series of a group $G$. Then $\gamma_{i+1}(G) \subseteq G_i$ for all $i$.*

Proof. We proceed by induction on $i$. The case $i = 0$ is trivial. So assume the result holds for some $i \geq 0$. Let $\pi_i \colon G \to G/G_i$ be the canonical map. Then

$$\gamma_{i+1}(G) = [G, \gamma_i(G)] \subseteq [G, G_{i-1}].$$

Since $\pi_i(G_{i-1}) \subseteq Z(G/G_i)$,

$$\pi_i([G, G_{i-1}]) = [\pi_i(G), \pi_i(G_{i-1})] = \{1\}.$$

Hence $\gamma_{i+1}(G) \subseteq [G, G_{i-1}] \subseteq G_i$. □

6.9. Theorem. *A group is nilpotent if and only if it admits a central series.*

Proof. Let $G$ be a group. If $G$ is nilpotent, then the $\gamma_j(G)$ form a central series of $G$. Conversely, if $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$ is a central series of $G$, then, by the previous lemma,

$$\gamma_{n+1}(G) \subseteq G_n = \{1\}.$$

Hence $G$ is nilpotent. □

6.10. Exercise. Let $G$ be a group. Prove that if $K$ is a subgroup of $Z(G)$ such that $G/K$ is nilpotent, then $G$ is nilpotent.

## 7. Hirsch's theorem

7.1. Theorem (Hirsch). *Let $G$ be a non-trivial nilpotent group. If $H$ is a non-trivial normal subgroup of $G$, then $H \cap Z(G) \neq \{1\}$. In particular, $Z(G) \neq \{1\}$.*

Proof. Since $\zeta_0(G) = \{1\}$ and there exists an integer $c$ such that $\zeta_c(G) = G$, there exists

$$m = \min\{k : H \cap \zeta_k(G) \neq \{1\}\}.$$

Since $H$ is normal in $G$,

$$[G, H \cap \zeta_m(G)] \subseteq H \cap [G, \zeta_m(G)] \subseteq H \cap \zeta_{m-1}(G) = \{1\}.$$

Therefore $\{1\} \neq H \cap \zeta_m(G) \subseteq H \cap Z(G)$. If $H = G$, then $Z(G) \neq \{1\}$. □

7.2. Exercise. Let $G$ be a nilpotent group and $M$ be a minimal normal subgroup of $G$. Prove that $M \subseteq Z(G)$.

7.3. Definition. Let $G$ be a group. A subgroup $M$ is said to be **maximal normal** in $G$ if $M \neq G$ and $M$ is the only proper normal subgroup of $G$ containing $M$.

7.4. Corollary. *Let $G$ be a non-abelian nilpotent group, and $A$ be a maximal normal and abelian subgroup of $G$. Then $A = C_G(A)$.*

Proof. Since $A$ is abelian, $A \subseteq C_G(A)$. Assume that $A \neq C_G(A)$. The centralizer $C_G(A)$ is normal in $G$, since $A$ is normal in $G$ and

$$gC_G(A)g^{-1} = C_G(gAg^{-1}) = C_G(A)$$

for all $g \in G$. Let $\pi\colon G \to G/A$ be the canonical map. Then $\pi(C_G(A))$ is a non-trivial normal subgroup of $\pi(G)$. Since $G$ is nilpotent, $\pi(G)$ is nilpotent. By Hirsch's theorem,

$$\pi(C_G(A)) \cap Z(\pi(G)) \neq \{1\}.$$

Let $x \in C_G(A) \setminus A$ be such that $\pi(x)$ is central in $\pi(G)$. Then $\langle A, x\rangle$ is abelian, as $x \in C_G(A)$. Moreover, $\langle A, x\rangle$ is normal in $G$, as $A$ is normal in $G$ and $gxg^{-1}x^{-1} \in A$ for all $g \in G$ (because $\pi(x)$ is central). Hence $gxg^{-1} \in \langle A, x\rangle$ and therefore $A \subsetneq \langle A, x\rangle \subsetneq G$, a contradiction. □

7.5. Theorem. *Let $G$ be a nilpotent group. The following statements hold:*

**1)** *Every minimal normal subgroup of $G$ has prime order and is central.*
**2)** *Every maximal subgroup of $G$ is normal of prime index and contains $[G, G]$.*

Proof.

**1)** Let $N$ be a minimal normal subgroup of $G$. Since $N \cap Z(G) \neq \{1\}$ by Hirsch's theorem, $N \cap Z(G)$ is a normal subgroup of $G$ contained in $N$. Then $N = N \cap Z(G) \subseteq Z(G)$ by the minimality of $N$. In particular, $N$ is abelian. Since every subgroup of $N$ is normal in $G$, $N$ is simple. Hence $N \simeq C_p$ for some prime number $p$.

**2)** If $M$ is a maximal subgroup, then $M$ is normal in $G$ by the normalizer condition (Lemma 5.18). By the maximality of $M$, the quotient $G/M$ contains no proper non-trivial subgroups. Thus $G/M \simeq C_p$ for some prime $p$. Since $G/M$ is abelian, $[G, G] \subseteq M$. □

The previous theorem does not prove the existence of maximal subgroups. For example, $\mathbb{Q}$ is a nilpotent group (as it is abelian) that contains no maximal subgroups.

7.6. Proposition. *Let $G$ be a nilpotent group and $H$ be a subgroup with $(G : H) = n$. If $g \in G$, then $g^n \in H$.*

Proof. We proceed by induction on $n$. The case $n = 1$ is trivial. The case $n = 2$ follows from the normality of $H$. So assume the result holds for all groups of index $< n$. Let $H$ be a subgroup of $G$ such that $(G : H) = n$. Let $H_0 = H$ and $H_{i+1} = N_G(H_i)$ for all $i \geq 0$. By definition, $H_i$ is normal in $H_{i+1}$. Since $G$ is nilpotent, $H_i \neq G$ implies that $H_i \subsetneq H_{i+1}$ by the normalizer condition. Since $(G : H)$ is finite, there exists $k$ such that $H_k = G$. Since $(H_j : H_{j-1}) < n$ for all $j$, the inductive hypothesis implies that $x^{(H_j:H_{j-1})} \in H_{j-1}$ for all $x \in H_j$ and all $j$. Hence

$$g^{(G:H)} = g^{(H_k:H_{k-1})(H_{k-1}:H_{k-2})\cdots(H_1:H_0)} \in H. \qquad \square$$

7.7. Exercise. Does the previous proposition hold for non-nilpotent groups?

The following lemma is useful for performing induction on the nilpotency index of nilpotent groups.

7.8. Lemma. *Let $G$ be a nilpotent group of class $c \geq 2$. If $x \in G$, then the subgroup $\langle x, [G, G]\rangle$ is nilpotent of class $< c$.*

Proof. Let $H = \langle x, [G, G]\rangle$. If $x \in [G, G]$, then there is nothing to prove. So assume that $x \notin [G, G]$. Note that

$$H = \{x^n c : n \in \mathbb{Z}, c \in [G, G]\},$$

as $[G, G]$ is normal in $G$. We need to show that $[H, H] \subseteq \gamma_3(G)$. Let $h = x^n c, k = x^m d \in H$ be such that $c, d \in [G, G]$. Since

$$[h, x^m] = [x^n, [c, x^m]][c, x^m] \in \gamma_4(G)\gamma_3(G) \subseteq \gamma_3(G),$$

then

$$\begin{aligned}[h, k] &= [h, x^m][x^m, [h, d]][h, d] \\ &= [x^n, [c, x^m]][c, x^m][x^m, [h, d]][h, d] \in \gamma_3(G).\end{aligned} \qquad \square$$

7.9. Example. Let $G = \mathbb{D}_8 = \langle r, s : r^8 = s^2 = 1, srs = r^{-1}\rangle$ the dihedral group of order 16. Then $G$ is nilpotent of class three and $[G, G] = \{1, r^2, r^4, r^6\} \simeq C_4$. The subgroup $\langle s, [G, G]\rangle \simeq \mathbb{D}_4$ is nilpotent of class two.

```
gap> G := DihedralGroup(IsPermGroup,16);;
gap> gens := GeneratorsOfGroup(G);;
gap> r := gens[1];;
gap> s := gens[2];;
gap> D := DerivedSubgroup(G);;
gap> S := Subgroup(G, Concatenation(Elements(D), [s]));;
gap> StructureDescription(S);
"D8"
gap> NilpotencyClassOfGroup(G);
3
gap> NilpotencyClassOfGroup(S);
2
```

Let us discuss a concrete application of Lemma 7.8.

7.10. Theorem. *If $G$ is a nilpotent group, then*

$$T(G) = \{g \in G : g^n = 1 \text{ for some } n \geq 1\}$$

*is a subgroup of $G$.*

Proof. We proceed by induction on the nilpotency class of $G$. Let $a, b \in T(G)$ and

$$A = \langle a, [G,G]\rangle, \quad B = \langle b, [G,G]\rangle.$$

Since $A$ and $B$ are nilpotent of class $< c$ by the previous lemma, the inductive hypothesis implies that $T(A)$ is a subgroup of $A$ and $T(B)$ is a subgroup of $B$. Since $T(A)$ is characteristic in $A$ and $A$ is normal in $G$, $T(A)$ is normal in $G$. Similarly, $T(B)$ is normal in $G$.

We claim that every element of $T(A)T(B)$ has finite order. If $x \in T(A)T(B)$, say $x = a_1b_1$ with $a_1$ of order $m$, then $x$ has finite order, as

$$x^m = (a_1b_1)^m = (a_1b_1a_1^{-1})(a_1^2b_1a_1^{-2})\cdots(a_1^{m-1}b_1a_1^{-m+1})b_1 \in T(B).$$

To see clearly what is what we did, let us work out a concrete example, say $m = 3$. In this case, we obtain the following formula:

$$\begin{aligned}(a_1b_1)^3 &= (a_1b_1)(a_1b_1)(a_1b_1)\\ &= (a_1b_1a_1^{-1})(a_1^2b_1a_1^{-2})a_1^3b_1 = (a_1b_1a_1^{-1})(a_1^2b_1a_1^{-2})b_1,\end{aligned}$$

as $a_1^3 = 1$.

With this trick, we prove that $ab$ has finite order. Hence $T(G)$ is a subgroup of $G$. □

Another application:

7.11. Theorem. *Let $G$ be a torsion-free nilpotent group and $a, b \in G$. If there exists $n \neq 0$ such that $a^n = b^n$, then $a = b$.*

Proof. We proceed by induction on the nilpotency order $c$ of $G$. The result clearly holds for abelian groups. Assume that $G$ is nilpotent of class $c \geq 2$. Since $\langle a, [G,G]\rangle$ is a nilpotent subgroup of $G$ of class $< c$ and $bab^{-1} = [b,a]a \in \langle a, [G,G]\rangle$, the inductive hypothesis implies that $ba = ab$, as

$$a^n = (bab^{-1})^n = b^n.$$

Thus $(ab^{-1})^n = a^nb^{-n} = 1$. Since $G$ has no torsion, we conclude that $a = b$. □

7.12. Corollary. *Let $G$ be a torsion-free nilpotent group. If $x, y \in G$ are such that $x^n y^m = y^m x^n$ for some $n, m \neq 0$, then $xy = yx$.*

Proof. Let $a = x$ and $b = y^n x y^{-n}$. Since $a^m = b^m$, the previous theorem implies that $a = b$. Thus $xy^n = y^n x$. Apply the previous theorem again, this time with $a = y$ and $b = xyx^{-1}$. Then we conclude that $xy = yx$. □

Before proving another theorem, we recall a basic lemma about finitely generated groups.

7.13. Lemma. *Let $G$ be a finitely generated group and $H$ a finite-index subgroup. Then $H$ is finitely generated.*

Proof. Assume that $G$ is generated by $\{g_1, \ldots, g_m\}$. Without loss of generality, we may assume that for each $i$ there exists $k$ such that $g_i^{-1} = g_k$.

Let $\{1 = t_1, \ldots, t_n\}$ be a right transversal of $H$ in $G$, so $G$ decomposes as

$$G = \bigcup_{i=1}^{n} H t_i \quad \text{(disjoint union)}.$$

For $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, m\}$, write

$$t_i g_j = h(i,j) t_{k(i,j)}.$$

We claim that $H$ is generated by the $h(i,j)$. Let $x \in H$. Then

$$\begin{aligned} x &= g_{i_1} \cdots g_{i_s} \\ &= (t_1 g_{i_1}) g_{i_2} \cdots g_{i_s} \\ &= h(1, i_1) t_{k_1} g_{i_2} \cdots g_{i_s} \\ &= h(1, i_1) h(k_1, i_2) t_{k_2} g_{i_3} \cdots g_{i_s} \\ &= h(1, i_1) h(k_1, i_2) \cdots h(k_{s-1}, i_s) t_{k_s}, \end{aligned}$$

where $k_1, \ldots, k_{s-1} \in \{1, \ldots, n\}$. Since $t_{k_s} \in H$ (because $x \in H$), $t_{k_s} = 1 \in H$. Hence $x$ is generated by the $h(i,j)$. □

Now the theorem:

7.14. Theorem. *Let $G$ be a finitely generated torsion group that is nilpotent. Then $G$ is finite*

Proof. We proceed by induction on the nilpotency class $c$ of $G$. The case $c = 1$ is true, as $G$ is abelian. So assume the result holds for groups of class $c \geq 1$. Since $G/[G,G]$ is a finitely generated abelian torsion group, it is finite. Thus $[G,G]$ has finite index in $G$. By Lemma 7.13, $[G,G]$ is finitely generated. Moreover, $[G,G]$ is a finitely generated nilpotent torsion group of class $< c$. By the inductive hypothesis, $[G,G]$ is a finite group. Since $[G,G]$ and $G/[G,G]$ are finite, $G$ is finite. □

## 8. Finite nilpotent groups

Before studying finite nilpotent groups, we need a lemma.

8.1. Lemma. *Let $G$ be a finite group and $p$ a prime number dividing $|G|$. If $P \in \mathrm{Syl}_p(G)$, then*

$$N_G(N_G(P)) = N_G(P).$$

Proof. Let $H = N_G(P)$. Since $P$ is normal in $H$, $P$ is the only Sylow $p$-subgroup of $H$. To prove that $N_G(H) = H$, it is enough to see that $N_G(H) \subseteq H$. Let $g \in N_G(H)$. Since

$$gPg^{-1} \subseteq gHg^{-1} = H,$$

$gPg^{-1} \in \mathrm{Syl}_p(H)$ and $H$ has only one Sylow $p$-subgroup, $P = gPg^{-1}$. Hence $g \in N_G(P) = H$. □

8.2. Theorem. *Let $G$ be a finite group. The following statements are equivalent:*

**1)** *$G$ is nilpotent.*
**2)** *Every Sylow subgroup of $G$ is normal in $G$.*
**3)** *$G$ is a direct product of its Sylow subgroups.*

Proof. We first prove that (1) $\implies$ (2). Let $P \in \mathrm{Syl}_p(G)$. We aim to prove that $P$ is normal in $G$, that is $N_G(P) = G$. By Lemma 8.1, $N_G(N_G(P)) = N_G(P)$. Now the normalizer condition (Lemma 5.18) implies that $N_G(P) = G$.

We now prove that (2) $\implies$ (3). Let $p_1, \cdots, p_k$ be the prime factors of $|G|$. For each $i \in \{1, \ldots, k\}$, let $P_i \in \mathrm{Syl}_{p_i}(G)$. By assumption, each $P_j$ is normal in $G$.

We claim that $P_1 \cdots P_j \simeq P_1 \times \cdots \times P_j$ for all $j$. The case $j = 1$ is trivial. So assume the result holds for some $j \geq 1$. Since

$$N = P_1 \cdots P_j \simeq P_1 \times \cdots \times P_j$$

is normal in $G$ and it has order coprime with $|P_{j+1}|$,

$$N \cap P_{j+1} = \{1\}.$$

Hence

$$NP_{j+1} \simeq N \times P_{j+1} \simeq P_1 \times \cdots \times P_j \times P_{j+1},$$

as $P_{j+1}$ is normal in $G$. Since now $P_1 \cdots P_k \simeq P_1 \times \cdots \times P_k$ is a subgroup of $G$ of order $|G|$, we conclude that $G = P_1 \times \cdots \times P_k$.

Finally, we prove that (3) $\implies$ (1). We just need to note that every $p$-group is nilpotent (Proposition 5.13) and that the direct product of nilpotent groups is nilpotent. □

8.3. Exercise. Let $G$ be a finite group. Prove that if $P \in \mathrm{Syl}_p(G)$ and $M$ is a subgroup of $G$ such that $N_G(P) \subseteq M$, then $M = N_G(M)$.

8.4. Exercise. Let $G$ be a finite group. Prove that the following statements are equivalent:

**1)** $G$ is nilpotent.
**2)** If $H \subsetneq G$ is a subgroup of $G$, then $H \subsetneq N_G(H)$.
**3)** Every maximal subgroup of $G$ is normal in $G$.

8.5. Theorem. *Let $G$ be a finite nilpotent group. If $p$ is a prime number dividing $|G|$, there exist a minimal normal subgroup of order $p$ and there exists a maximal subgroup of index $p$.*

Proof. Assume that $|G| = p^\alpha m$ with $\gcd(p, m) = 1$. Write $G = P \times H$, where $P \in \mathrm{Syl}_p(G)$. Since $Z(P)$ is a non-trivial normal subgroup of $P$, every subgroup of $Z(P)$ that is minimal normal in $G$ has order $p$ (and such subgroups exist because $G$ is finite). Since $P$ contains a subgroup of index $p$, it is maximal. Hence $P \times H$ contains a maximal subgroup of index $p$. □

8.6. Exercise. Let $p$ be a prime number and $G$ be a non-trivial group of order $p^n$. Prove the following statements:

**1)** $G$ has a normal subgroup of order $p$.
**2)** For every $j \in \{0, \dots, n\}$ there exists a normal subgroup of $G$ of order $p^j$.

8.7. Exercise. Let $G$ be a finite group. Prove that the following statements are equivalent:

**1)** $G$ is nilpotent.
**2)** Any two elements of coprime order commute.
**3)** Every non-trivial quotient of $G$ has a non-trivial center.
**4)** If $d$ divides $|G|$, then there exists a normal subgroup of $G$ of order $d$.

## 9. The Baumslag–Wiegold theorem

The following result can be proved with elementary tools and was discovered in 2014.

9.1. Theorem (Baumslag–Wiegold). *Let $G$ be a finite group such that $|xy| = |x||y|$ for all $x, y \in G$ of coprime orders. Then $G$ is nilpotent.*

Proof. Let $p_1, \dots, p_n$ be the prime factors of $|G|$. For each $i \in \{1, \dots, n\}$, let $P_i \in \mathrm{Syl}_{p_i}(G)$. We first prove that $G = P_1 \cdots P_n$. To prove the non-trivial inclusion, we need to show that the map

$$\psi \colon P_1 \times \cdots \times P_n \to G, \quad (x_1, \dots, x_n) \mapsto x_1 \cdots x_n$$

is surjective. We first show that $\psi$ is injective: If $\psi(x_1, \dots, x_n) = \psi(y_1, \dots, y_n)$, then

$$x_1 \cdots x_n = y_1 \cdots y_n.$$

If $y_n \neq x_n$, then $x_1 \cdots x_{n-1} = (y_1 \cdots y_{n-1}) y_n x_n^{-1}$. Since $x_1 \cdots x_{n-1}$ has order coprime with $p_n$ and $y_1 \cdots y_{n-1} y_n x_n^{-1}$ has order a multiple of $p_n$, we get a contradiction. Thus $x_n = y_n$. The same argument shows that $\psi$ is injective. Since $|P_1 \times \cdots \times P_n| = |G|$, we conclude that $\psi$ is bijective. In particular, $\psi$ is surjective.

We now prove that each $P_j$ is normal in $G$. Let $j \in \{1, \dots, n\}$ and $x_j \in P_j$. Let $g \in G$ and $y_j = g x_j g^{-1}$. Since $y_j \in G$, we can write $y_j = z_1 \cdots z_n$ with $z_k \in P_k$ for all $k$. Since the order of $y_j$ is a power of $p_j$, the element $z_1 \cdots z_n$ has order a power of $p_j$. Thus $z_k = 1$ for all $k \neq j$. Moreover, $y_j = z_j \in P_j$. Since every Sylow subgroup of $G$ is normal in $G$, we conclude that $G$ is nilpotent. □

## 10. Frattini subgroup

10.1. Definition. Let $G$ be a group. If $G$ has maximal subgroups, the **Frattini subgroup** is the intersection $\Phi(G)$ of all the maximal subgroups of $G$. Otherwise, $\Phi(G) = G$.

10.2. Exercise. Prove that $\Phi(G)$ is a characteristic subgroup of $G$.

10.3. Example. Let $G = \mathbb{S}_3$. The maximal subgroups of $G$ are

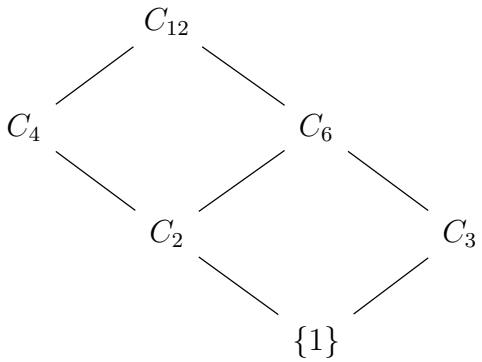
$$M_1 = \langle (123) \rangle, \quad M_2 = \langle (12) \rangle, \quad M_3 = \langle (23) \rangle, \quad M_4 = \langle (13) \rangle.$$

Hence $\Phi(G) = \{1\}$.

10.4. Example. Let $G = \langle g \rangle \simeq C_{12}$. The subgroups of $G$ are

$$\{1\}, \quad \langle g^6 \rangle \simeq C_2, \quad \langle g^4 \rangle \simeq C_3, \quad \langle g^3 \rangle \simeq C_4, \quad \langle g^2 \rangle \simeq C_6, \quad G.$$

Let us draw a picture:




The maximal subgroups of $G$ are $\langle g^3 \rangle \simeq C_4$ and $\langle g^2 \rangle \simeq C_6$. Hence $\Phi(G) = \langle g^3 \rangle \cap \langle g^2 \rangle = \langle g^6 \rangle \simeq C_2$. Let us see how to do this calculation with the computer:

```
gap> G = CyclicGroup(12);;
gap> StructureDescription(FrattiniSubgroup(G));
"C2"
```

10.5. Lemma (Dedekind). *Let $H$, $K$ and $L$ be subgroups of $G$ such that $H \subseteq L \subseteq G$. Then $HK \cap L = H(K \cap L)$.*

Proof. One inclusion is trivial. Let us prove then that $HK \cap L \subseteq H(K \cap L)$. If $x = hk \in HK \cap L$ with $x \in L$, $h \in H$ and $k \in K$, then $k = h^{-1}x \in L \cap K$, as $H \subseteq L$. Thus $x = hk \in H(L \cap K)$. □

10.6. Lemma. *Let $G$ be a finite group and $H$ be a subgroup of $G$ such that $G = H\Phi(G)$. Then $H = G$.*

Proof. If $H \neq G$, let $M$ be a maximal subgroup of $G$ such that $H \subseteq M$. Since $\Phi(G) \subseteq M$, $G = H\Phi(G) \subseteq M$, a contradiction. □

10.7. Proposition. *Let $N$ be a normal subgroup of a finite group $G$. Then $\Phi(N) \subseteq \Phi(G)$.*

PROOF. Since $\Phi(N)$ is characteristic in $N$ and $N$ is normal in $G$, $\Phi(N)$ is normal in $G$. If there exists a maximal subgroup $M$ such that $\Phi(N) \not\subseteq M$, then $\Phi(N)M = G$. (This happens because, otherwise, $M = \Phi(N)M \supseteq \Phi(N)$.) By Dedekind's lemma (with $H = \Phi(N)$, $K = M$ and $L = N$),

$$N = G \cap N = (\Phi(N)M) \cap N = \Phi(N)(M \cap N).$$

By Lemma 10.6 (with $G = N$ and $H = M \cap N$), $\Phi(N) \subseteq N \subseteq M$, a contradiction. Hence every maximal subgroup of $G$ contains $\Phi(N)$ and therefore $\Phi(G) \supseteq \Phi(N)$. □

The following proposition states that the elements of the Frattini subgroup are the **non-generators** of the group.

10.8. PROPOSITION. *Let $G$ be a finite group. Then*

$$\Phi(G) = \{x \in G : \text{if } G = \langle x, Y\rangle \text{ for some } Y \subseteq G, \text{ then } G = \langle Y\rangle\}.$$

PROOF. We first prove $\supseteq$. Let $x \in \{x \in G : \text{if } G = \langle x, Y\rangle \text{ for some } Y \subseteq G, \text{ then } G = \langle Y\rangle\}$. If $M$ is a maximal subgroup of $G$ such that $x \notin M$, then, since $G = \langle x, M\rangle$, we obtain that $G = \langle M\rangle = M$, a contradiction. Thus $x \in M$ for all maximal subgroup $M$ of $G$. Hence $x \in \Phi(G)$.

We now prove $\subseteq$. Let $x \in \Phi(G)$ be such that $G = \langle x, Y\rangle$ for some subset $Y$ of $G$. If $G \neq \langle Y\rangle$, there exists a maximal subgroup $M$ such that $\langle Y\rangle \subseteq M$. Since $x \in M$, $G = \langle x, Y\rangle \subseteq M$, a contradiction. □

10.9. EXAMPLE. For a prime number $p$, let $G$ be an elementary $p$-group, that is $G \simeq C_p^m$ for some $m \geq 1$. Assume that $G = \langle x_1\rangle \times \cdots \times \langle x_m\rangle$ with $\langle x_j\rangle \simeq C_p$. We claim that $\Phi(G)$ is trivial. For $j \in \{1, \ldots, m\}$, let $n_j \in \{1, \ldots, p-1\}$. Since

$$\{x_1, \ldots, x_{j-1}, x_j^{n_j}, x_{j+1}, \ldots, x_m\}$$

generates $G$ and $\{x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_m\}$ does not, $x_j^{n_j} \notin \Phi(G)$ by Proposition 10.8. Hence $\Phi(G) = \{1\}$.

10.10. THEOREM (Frattini). *Let $G$ be a finite group. Then $\Phi(G)$ is nilpotent.*

PROOF. Let $P \in \mathrm{Syl}_p(\Phi(G))$ for some prime number $p$. Since $\Phi(G)$ is normal in $G$, Lemma 2.1 (Frattini's argument) implies that $G = \Phi(G)N_G(P)$. By Lemma 10.6, $G = N_G(P)$. Since every Sylow subgroup of $\Phi(G)$ is normal in $G$, $\Phi(G)$ is nilpotent. □

10.11. EXERCISE. Let $G$ be a group and $M$ be a normal subgroup of $G$. Prove that if $M$ is maximal, then $G/M$ is cyclic of prime order.

10.12. THEOREM (Gaschütz). *If $G$ is a finite group, then*

$$[G, G] \cap Z(G) \subseteq \Phi(G).$$

PROOF. Let $D = [G, G] \cap Z(G)$. Assume that $D$ is not contained in $\Phi(G)$. Since $\Phi(G)$ is contained in every maximal subgroup of $G$, there is a maximal subgroup $M$ of $G$ not containing $D$. Then $G = MD$. Since $D \subseteq Z(G)$, $M$ is normal in $G$, as $g = md \in G = MD$ implies

$$gMg^{-1} = (md)Md^{-1}m^{-1} = mMm^{-1} = M.$$

Since $G/M$ is cyclic of prime order, $G/M$ is, in particular, abelian and hence $[G, G] \subseteq M$. Therefore $D \subseteq [G, G] \subseteq M$, a contradiction. □

10.13. Lemma. *Let $G$ be a finite group and $P \in \mathrm{Syl}_p(G)$. If $H$ is a subgroup of $G$ such that $N_G(P) \subseteq H$, then $N_G(H) = H$.*

Proof. Let $x \in N_G(H)$. Since $P \in \mathrm{Syl}_p(H)$ and $Q = xPx^{-1} \in \mathrm{Syl}_p(H)$, the second Sylow's theorem implies that there exists $h \in H$ such that $hQh^{-1} = (hx)P(hx)^{-1} = P$. Then $hx \in N_G(P) \subseteq H$ and hence $x \in H$. □

10.14. Theorem (Wielandt). *A finite group $G$ is nilpotent if and only if $[G, G] \subseteq \Phi(G)$.*

Proof. Assume that $[G, G] \subseteq \Phi(G)$. Let $P \in \mathrm{Syl}_p(G)$. If $N_G(P) \neq G$, then $N_G(P) \subseteq M$ for some maximal subgroup $M$ of $G$. If $g \in G$ and $m \in M$, then, since

$$gmg^{-1}m^{-1} = [g, m] \in [G, G] \subseteq \Phi(G) \subseteq M,$$

$M$ is normal in $G$. Furthermore $N_G(P) \subseteq M$. By Lemma 10.13,

$$G = N_G(M) = M,$$

a contradiction. Thus $N_G(P) = G$ and every Sylow subgroup of $G$ si normal in $G$. Therefore $G$ is nilpotent.

Conversely, assume that $G$ is nilpotent. Let $M$ be a maximal subgroup of $G$. Since $M$ is normal in $G$ and maximal, $G/M$ has no proper non-trivial subgroups. Then $G/M \simeq C_p$ for some prime number $p$. In particular, $G/M$ is abelian and $[G, G] \subseteq M$. Since $[G, G]$ is contained in every maximal subgroup of $G$, $[G, G] \subseteq \Phi(G)$. □

10.15. Theorem. *A finite group $G$ is nilpotent if and only if $G/\Phi(G)$ is nilpotent.*

Proof. If $G$ is nilpotent, then $G/\Phi(G)$ is nilpotent. Conversely, assume that $G/\Phi(G)$ is nilpotent. Let $P \in \mathrm{Syl}_p(G)$. Since $\Phi(G)P/\Phi(G) \in \mathrm{Syl}_p(G/\Phi(G))$ and $G/\Phi(G)$ is nilpotent, $\Phi(G)P/\Phi(G)$ is a normal subgroup of $G/\Phi(G)$. By the correspondence theorem, $\Phi(G)P$ is a normal subgroup of $G$. Since $P \in \mathrm{Syl}_p(\Phi(G)P)$, Frattini's argument (Lemma 2.1) implies that

$$G = \Phi(G)PN_G(P) = \Phi(G)N_G(P),$$

as $P \subseteq N_G(P)$. Thus $G = N_G(P)$ by Lemma 10.6). Hence $P$ is normal in $G$ and therefore $G$ is nilpotent. □

10.16. Theorem (Hall). *Let $G$ be a finite group with a normal subgroup $N$. If both $N$ and $G/[N,N]$ are nilpotent, then $G$ is nilpotent.*

Proof. Since $N$ is nilpotent, $[N,N] \subseteq \Phi(N)$ by Wielandt's theorem 10.14. By Proposition 10.7, $[N,N] \subseteq \Phi(N) \subseteq \Phi(G)$. By the universal property, there exists a surjective group homomorphism $G/[N,N] \to G/\Phi(G)$ such that the diagram

$$\begin{array}{ccc} G & \longrightarrow & G/\Phi(G) \\ \downarrow & \nearrow & \\ G/[N,N] & & \end{array}$$

is commutative. Since $G/[N,N]$ is nilpotent, $G/\Phi(G)$ is nilpotent. Thus $G$ is nilpotent by the previous theorem. □

10.17. Definition. A **minimal generating set** of a group $G$ is a set $X$ of generators of $G$ such that no proper subset of $X$ generates $G$.

Note that a minimal generating set does not necessarily have minimal size.

10.18. Example. Let $G = \langle g \rangle \simeq C_6$. If $a = g^2$ and $b = g^3$, then $\{a, b\}$ is a minimal generating set of $G$ that does not have minimal size, as $G = \langle ab \rangle$.

For a prime number $p$, we write $\mathbb{F}_p$ to denote the field of $p$ elements.

10.19. Lemma. *Let $p$ be a prime number and $G$ be a finite $p$-group. Then $G/\Phi(G)$ is a vector space over $\mathbb{F}_p$.*

Proof. Let $K$ be a maximal subgroup of $G$. Since $G$ is nilpotent (see Proposition 5.13), $K$ is normal in $G$ (Exercise 8.4). Thus $G/K \simeq C_p$ because it is a simple $p$-group.

It is enough to prove that $G/\Phi(G)$ is an elementary abelian $p$-group. It is a $p$-group because $G$ is a $p$-group. Let $K_1, \ldots, K_m$ be the maximal subgroups of $G$. If $x \in G$, then $x^p \in K_j$ for all $j \in \{1, \ldots, m\}$. Hence $x^p \in \Phi(G) = \cap_{j=1}^m K_j$. Moreover, $G/\Phi(G)$ is abelian, as $[G,G] \subseteq \Phi(G)$ because $G$ is nilpotent (Wielandt's theorem 10.14). □

10.20. Theorem (Burnside). *Let $p$ be a prime number and $G$ a finite $p$-group. If $X$ is a minimal generating set of $G$, then $|X| = \dim G/\Phi(G)$.*

Proof. By Lemma 10.19, $G/\Phi(G)$ is a vector space over $\mathbb{F}_p$. Let $\pi\colon G \to G/\Phi(G)$ be the canonical map and $\{x_1, \ldots, x_n\}$ be a minimal generating set of $G$. We claim that $\{\pi(x_1), \ldots, \pi(x_n)\}$ is a linearly independent subset of $G/\Phi(G)$. Assume this is not the case. Without loss of generality, let us assume that $\pi(x_1) \in \langle \pi(x_2), \ldots, \pi(x_n) \rangle$. There exists $y \in \langle x_2, \ldots, x_n \rangle$ such that $x_1 y^{-1} \in \Phi(G)$. Since $G$ is generated by $\{x_1 y^{-1}, x_2, \ldots, x_n\}$ and $x_1 y^{-1} \in \Phi(G)$, Proposition 10.8 implies that $G$ is generated by $\{x_2, \ldots, x_n\}$, a contradiction to the minimality of $\{x_1, \ldots, x_n\}$. Therefore $n = \dim G/\Phi(G)$. □

10.21. Exercise. Let $p$ be a prime number and $G$ a finite $p$-group. Prove that if $x \notin \Phi(G)$, then $x$ belongs to some minimal set of generators of $G$.

## 11. The Fitting subgroup

11.1. Definition. Let $G$ be a finite group and $p$ be a prime number. The **$p$-radical** of $G$ is the subgroup

$$O_p(G) = \bigcap_{P \in \mathrm{Syl}_p(G)} P.$$

11.2. Lemma. *Let $G$ be a finite group and $p$ be a prime number. The following statements hold:*

**1)** *$O_p(G)$ is normal in $G$.*
**2)** *If $N$ is a normal subgroup of $G$ contained in some $P \in \mathrm{Syl}_p(G)$, then $N \subseteq O_p(G)$.*

Proof. Let $P \in \mathrm{Syl}_p(G)$. Let $G$ act on $G/P$ by left multiplication. There is a group homomorphism $\rho\colon G \to \mathbb{S}_{G/P}$ with kernel

$$\begin{aligned}\ker \rho &= \{x \in G : \rho_x = \mathrm{id}\} \\ &= \{x \in G : xgP = gP \ \forall g \in G\} \\ &= \{x \in G : x \in gPg^{-1} \ \forall g \in G\} \\ &= \bigcap_{g \in G} gPg^{-1} \\ &= O_p(G).\end{aligned}$$

Then $O_p(G)$ is normal in $G$.

Let $N$ be a normal subgroup of $G$ such that $N \subseteq P$. Since $N = gNg^{-1} \subseteq gPg^{-1}$ for all $g \in G$, we conclude that $N \subseteq O_p(G)$. □

11.3. Definition. Let $G$ be a finite group and $p_1, \dots, p_k$ be the prime divisors of $|G|$. The **Fitting subgroup** of $G$ is the subgroup

$$F(G) = O_{p_1}(G) \cdots O_{p_k}(G)$$

11.4. Exercise. Prove that $F(G)$ is characteristic in $G$.

11.5. Example. Let $G = \mathbb{S}_3$. Then $O_2(G) = \{1\}$ and $O_3(G) = \langle (123) \rangle$. Hence $F(G) = \langle (123) \rangle$.

11.6. Theorem (Fitting). *Let $G$ be a finite group. Then $F(G)$ is a nilpotent and normal in $G$. Moreover, $F(G)$ contains every nilpotent normal subgroup of $G$.*

Proof. By definition, $|F(G)| = \prod_p |O_p(G)|$. Since $O_p(G) \in \mathrm{Syl}_p(F(G))$, we conclude that $F(G)$ is nilpotent, as it contains a normal Sylow $p$-subgroup for every prime $p$. Hence $F(G)$ is nilpotent by Theorem 8.2.

Let $N$ be a nilpotent normal subgroup of $G$ and $P \in \mathrm{Syl}_p(N)$. Since $N$ is nilpotent, $P$ is normal in $N$ and hence $P$ is the only Sylow $p$-subgroup of $N$. Thus $P$ is characteristic in $N$ and $P$ is normal in $G$. Since $N$ is nilpotent, $N$ is a direct product of its Sylow subgroups. Therefore $N \subseteq O_p(G)$ by Lemma 11.2. □

11.7. Corollary. *If $G$ is a finite group, then $Z(G) \subseteq F(G)$.*

Proof. Since $Z(G)$ is nilpotent (in fact, it is abelian) and normal in $G$, by Fitting's theorem 11.6 we conclude that $Z(G) \subseteq F(G)$ □

11.8. Corollary (Fitting). *Let $K$ and $L$ be nilpotent normal subgroups of a finite group $G$. Then $KL$ is nilpotent.*

Proof. By Fitting's theorem 11.6, $K \subseteq F(G)$ and $L \subseteq F(G)$. Then $KL \subseteq F(G)$ and $KL$ is nilpotent, as $F(G)$ is nilpotent. □

11.9. Corollary. *Let $N$ be a normal subgroup of a finite group $G$. Then $N \cap F(G) = F(N)$.*

Proof. Since $F(N)$ is characteristic in $N$, $F(N)$ is normal in $G$. Then $F(N) \subseteq N \cap F(G)$ because $F(N)$ is nilpotent. Conversely, since $F(G)$ is normal in $G$, the subgroup $F(G) \cap N$ is normal in $N$. Since $F(G) \cap N$ is nilpotent, $F(G) \cap N \subseteq F(N)$. □

We now discuss an application to finite solvable groups.

11.10. Theorem. *Let $G$ be a non-trivial solvable group. Every normal non-trivial subgroup $N$ contains a normal abelian non-trivial subgroup. Moreover, this subgroup is contained in $F(N)$.*

Proof. Note that $N \cap G^{(0)} = N \neq \{1\}$. Since $G$ is solvable, there exists $m \geq 1$ such that $N \cap G^{(m)} = \{1\}$. Let $n$ be the largest positive integer such that $N \cap G^{(n)} \neq \{1\}$. Since $[N, N] \subseteq N$ and $[G^{(n)}, G^{(n)}] = G^{(n+1)}$,

$$[N \cap G^{(n)}, N \cap G^{(n)}] \subseteq N \cap G^{(n+1)} = \{1\}.$$

Then $N \cap G^{(n)}$ is an abelian subgroup of $G$. Moreover, it is normal and nilpotent. Hence

$$N \cap G^{(n)} \subseteq N \cap F(G) = F(N). \qquad \square$$

11.11. Theorem. *If $N$ is a minimal normal subgroup of a finite group $G$, then $F(G) \subseteq C_G(N)$.*

Proof. By Fitting's theorem 11.6, $F(G)$ is a normal nilpotent group. The subgroup $N \cap F(G)$ is normal in $G$. Moreover, $[F(G), N] \subseteq N \cap F(G)$. If $N \cap F(G) = \{1\}$, then $[F(G), N] = \{1\}$. Otherwise, $N = N \cap F(G) \subseteq F(G)$ by the minimality of $N$. By Hirsch's theorem, $N \cap Z(F(G)) \neq \{1\}$ Since $Z(F(G))$ is characteristic in $F(G)$ and $F(G)$ is normal in $G$, $Z(F(G))$ is normal in $G$. Since $\{1\} \neq N \cap Z(F(G))$ is normal in $G$, the minimality of $N$ implies that $N = N \cap Z(F(G)) \subseteq Z(F(G))$. Hence $[F(G), N] = \{1\}$. □

11.12. Corollary. *Let $G$ be a finite solvable group. The following statements hold:*

**1)** *If $N$ is a minimal normal subgroup, then $N \subseteq Z(F(G))$.*
**2)** *If $H$ is a non-trivial normal subgroup, then $H \cap F(G) \neq \{1\}$.*

Proof. Let us prove the first claim. Since $N$ is a $p$-group by Lemma 1.20, $N$ is nilpotent and hence $N \subseteq F(G)$. Moreover, $F(G) \subseteq C_G(N)$ by the previous theorem. Therefore $N \subseteq Z(F(G))$.

Let us prove now the second claim. The subgroup $H$ contains a minimal normal subgroup $N$ and $N \subseteq F(G)$. Then $H \cap F(G) \neq \{1\}$. □

11.13. Theorem. *Let $G$ be a finite group. The following statements hold:*

**1)** *$\Phi(G) \subseteq F(G)$ and $Z(G) \subseteq F(G)$.*
**2)** *$F(G)/\Phi(G) \simeq F(G/\Phi(G))$.*

PROOF. Let us prove the first claim. Since $\Phi(G)$ is normal in $G$, nilpotent by Frattini's Theorem 10.10 and $F(G)$ contains every normal nilpotent subgroup of $G$, $\Phi(G) \subseteq F(G)$. Moreover, $Z(G)$ is normal in $G$ and nilpotent. Hence $Z(G) \subseteq F(G)$.

Let us prove the second claim. Let $\pi\colon G \to G/\Phi(G)$ be the canonical map. Since $F(G)$ is nilpotent, $\pi(F(G))$ is nilpotent. Hence

$$\pi(F(G)) \subseteq F(G/\Phi(G))$$

by Fitting's Theorem 11.6. Let $H = \pi^{-1}(F(G/\Phi(G)))$. By the correspondence theorem, $H$ is a normal subgroup of $G$ containing $\Phi(G)$. If $P \in \mathrm{Syl}_p(H)$, then $\pi(P) \in \mathrm{Syl}_p(\pi(H))$. In fact, $\pi(P) \simeq P/P \cap \Phi(G)$ is a $p$-group and $(\pi(H) : \pi(P))$ is coprime with $p$ because

$$(\pi(H) : \pi(P)) = \frac{|\pi(H)|}{|\pi(P)|} = \frac{|H/\Phi(G)|}{|P/P \cap \Phi(G)|} = \frac{(H : P)}{(\Phi(G) : P \cap \Phi(G))}$$

divides $(H : P)$, a number coprime with $p$. Since $\pi(H)$ is nilpotent, $\pi(P)$ is characteristic in $\pi(H)$. Then $\pi(P)$ is normal in $\pi(G) = G/\Phi(G)$ and $P\Phi(G) = \pi^{-1}(\pi(P))$ is normal in $G$. Since $P \in \mathrm{Syl}_p(P\Phi(G))$, Frattini's argument (Lemma 2.1) implies that $G = \Phi(G)N_G(P)$. Therefore $P$ is normal in $G$ by Lemma 10.6. Moreover, $P$ is normal in $H$. Since $P$ is nilpotent and normal in $G$, $P \subseteq F(G)$ by Fitting's theorem 11.6. Hence $H \subseteq F(G)$ and $F(G/\Phi(G)) = \pi(H) \subseteq \pi(F(G))$. □

## 12. Super-solvable groups

12.1. Definition. A group $G$ is said to be **super-solvable** if there exists a sequence

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$$

of normal subgroups of $G$ such that every quotient $G_{i-1}/G_i$ is cyclic.

In the previous definition, we do not require the group to be finite. Hence the quotients could be finite cyclic groups or isomorphic to $\mathbb{Z}$.

12.2. Example. The dihedral group $\mathbb{D}_n$ of order $2n$ is super-solvable, as

$$\mathbb{D}_n \supseteq \langle r \rangle \supseteq \{1\}$$

is a sequence of normal subgroups with cyclic factors.

Every super-solvable group is solvable. See Exercise 1.6.

12.3. Example. The alternating group $\mathbb{A}_4$ solvable but not super-solvable. The only proper non-trivial normal subgroup of $\mathbb{A}_4$ is

$$\{\mathrm{id}, (12)(34), (13)(24), (14)(23)\} \simeq C_2 \times C_2.$$

Thus $\mathbb{A}_4$ does not have a sequence of normal subgroups with cyclic factors.

12.4. Example. The group $\mathbf{SL}_2(3)$ is solvable but not super-solvable. Here is a computer verification:

```
gap> IsSolvable(SL(2,3));
true
gap> IsSupersolvable(SL(2,3));
false
```

12.5. Exercise. Prove the following statements:

**1)** Every subgroup of a super-solvable group is super-solvable.
**2)** Quotients of super-solvable groups are super-solvable.

12.6. Exercise. Prove that the direct product of super-solvable groups is super-solvable.

12.7. Exercise. Let $H$ and $K$ be normal subgroups of a group $G$ such that $G/K$ and $G/H$ are super-solvable. Prove that $G/H \cap K$ is super-solvable.

12.8. Exercise. Let $N$ be a cyclic normal subgroup of $G$. If $G/N$ is super-solvable, then $G$ is super-solvable.

12.9. Theorem. *Let $G$ be a super-solvable non-trivial group. Then $G$ admits a sequence*

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$$

*of normal subgroups such that every quotient $G_{i-1}/G_i$ is cyclic of prime order or isomorphic to $\mathbb{Z}$.*

PROOF. Let $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$ be a sequence of normal subgroups of $G$ such that every quotient $G_{i-1}/G_i$ is cyclic. Let $i \in \{1, \ldots, n\}$ be such that $G_{i-1}/G_i \simeq C_n$ for some non-prime $n$ and let $\pi\colon G_{i-1} \to G_{i-1}/G_i$ be the canonical map. Let $p$ be a prime divisor of $n$ and $H$ be a subgroup of $G$ such that $\pi(H)$ is a subgroup of $G_{i-1}/G_i$ of order $p$. By the correspondence theorem, $G_i \subseteq H \subseteq G_{i-1}$.

We claim that $H$ is normal in $G$. Let $g \in G$. Since $\pi(gHg^{-1})$ is a subgroup of order $p$ of the cyclic group $G_{i-1}/G_i$, $\pi(gHg^{-1}) = \pi(H)$. Then $gHg^{-1} = G_iH \subseteq H$ and hence $gHg^{-1} = H$.

Note that $H/G_i$ is cyclic of prime order, as

$$H/G_i = H/H \cap G_i \simeq \pi(H) \simeq C_p.$$

Moreover, $G_{i-1}/H$ is cyclic, as

$$G_{i-1}/H \simeq \frac{G_{i-1}/G_i}{H/G_i}$$

is the quotient of a cyclic group.

We have shown that by adding $H$ to our sequence of normal subgroups, we obtain a sequence with cyclic factors where $H/G_i$ is cylic of prime order. Repeating this procedure, we obtain the desired result. □

Let us discuss an immediate application.

12.10. COROLLARY. *A finite super-solvable group admits a sequence of normal subgroups where each quotient is cyclic of prime order.*

We now discuss other properties of super-solvable groups.

12.11. THEOREM. *Let $G$ be a super-solvable group. The following statement hold:*

**1)** *If $N$ is minimal normal in $G$, then $N \simeq C_p$ for some prime number $p$.*

**2)** *If $M$ is maximal in $G$, then $(G : M) = p$ for some prime number $p$.*

PROOF. Let us prove the first claim. Since $G$ is super-solvable, there exists a sequence

$$G = G_0 \supseteq G_1 \supseteq G_2 \supseteq \cdots \supseteq G_n = \{1\}$$

of normal subgroups with cyclic factors. Since each $G_i \cap N$ is a normal subgroup of $G$ contained in $N$, the minimality implies that each $G_i \cap N$ is either trivial or equal to $N$. Moreover, $N \cap G_0 = N$ and $N \cap G_n = \{1\}$. Let $j$ be the smallest positive integer such that $N \cap G_j = \{1\}$. Since $N \subseteq G_{j-1}$ (because $N \cap G_{j-1} = N$), the composition

$$N \hookrightarrow G_{j-1} \to G_{j-1}/G_j$$

is an injective group homomorphism, as its kernel is equal $N \cap G_j = \{1\}$. Thus $N$ is cyclic, as it is isomorphic to a subgroup of the cyclic group $G_{i-1}/G_i$. If $G_{i-1}/G_i \simeq \mathbb{Z}$, then $N \simeq \mathbb{Z}$, a contradiction to the fact that $N$ is minimal normal. (For example, $2\mathbb{Z}$ is characteristic subgroup of $\mathbb{Z}$ and hence it is normal in $G$. Thus $N$ is cyclic and finite. Hence $N \simeq C_p$.)

We now prove the second claim. Let $M$ be a maximal subgroup of $G$. If $M$ is normal in $G$, then $G/M$ does not contain non-trivial proper subgroups. Then $G/M \simeq C_p$ for some prime number $p$. Assume that $M$ is not normal in $G$. Let $H = \cap_{g \in G} gMg^{-1}$ and $\pi\colon G \to G/H$ be the canonical map. Since $\pi(M)$ is maximal in $\pi(G) = G/H$ and

$$(G : M) = (G/H : M/H) = (G/H : M/H \cap M) = (\pi(G) : \pi(M)),$$

we may assume that $M$ does not contain non-trivial normal subgroups of $G$ (if needed, we just replace $G$ by $G/H$). Since $G$ is super-solvable, there exists a sequence $G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{1\}$ of normal subgroups of $G$ with factors either cyclic of prime order or isomorphic to $\mathbb{Z}$. Let $N = G_{n-1}$. Since $N$ is cyclic, every subgroup of $N$ is characteristic in $N$ and hence normal in $G$. In particular, $M \cap N$ is normal in $G$ and therefore $M \cap N = \{1\}$. Since $M \subseteq NM \subseteq G$, the maximality of $M$ implies that either $M = NM$ or $G = NM$. Since $N \subseteq NM = M$ yields a contradiction, we conclude that $G = NM$.

If $N \simeq C_p$ for some prime $p$, then $(G : M) = p$ and the proof is complete. Assume that $N \simeq \mathbb{Z}$. Let $H$ be a proper non-trivial subgroup of $N$. Since $H$ is characteristic in $N$, $H$ is normal in $G$. Since $M \subseteq HM \subseteq NM = G$, the maximality of $M$ implies that either $HM = M$ or $HM = G$. Since $HM = M$ implies $H \subseteq M \cap N = \{1\}$, we may assume that $HM = G$. If $n \in N \setminus H$, then $n = hm$ for some $h \in H$ and $m \in M$. Then $h = n$, as $h^{-1}n \in N \cap M = \{1\}$, a contradiction.

□

12.12. Exercise. Let $G$ be a super-solvable group. Prove the following statements:

**1)** The commutator subgroup $[G, G]$ is nilpotent.

**2)** If $G$ is non-abelian, there exists a normal subgroup $N \neq G$ such that $Z(G) \subsetneq N$.

There are solvable groups with a non-nilpotent derived subgroup.

12.13. Example. The group $\mathbb{S}_4$ is solvable and $[\mathbb{S}_4, \mathbb{S}_4] = \mathbb{A}_4$ is not nilpotent.

12.14. Proposition. *Let $p$ be a prime number. Every finite $p$-group is super-solvable.*

Proof. Let $G$ be a minimal counterexample. We may assume that $|G| = p^n$ for some $n > 1$ (otherwise, if $n = 1$, then $G$ is trivially super-solvable). The group $G$ is nilpotent and contains a normal subgroup $N$ of order $p$. Moreover, since $|G/N| = p^{n-1}$, the group $G/N$ is super-solvable. Since $N$ is cyclic and $G/N$ is super-solvable, $G$ is super-solvable by Exercise 12.8. □

12.15. Exercise. Prove that finite nilpotent groups are super-solvable.

12.16. Theorem. *Super-solvable groups have maximal subgroups.*

Proof. We proceed by induction on the length of the super-solvable series. The claim holds for groups with a super-solvable series of length one, as in this case we are dealing with cyclic groups. So let $G$ be a group admitting a sequence

$$G = G_0 \supseteq \cdots \supseteq G_k = \{1\}$$

and suppose the theorem holds for super-solvable groups with super-solvable series of length $< k$. Note that $G_{k-1}$ is normal in $G$. Let $\pi\colon G \to G/G_{k-1}$ be the canonical map. The sequence

$$G/G_{k-1} = \pi(G) \supseteq \pi(G_1) \supseteq \cdots \supseteq \pi(G_{k-1}) = \{1\}$$

has length $< k$ and proves the super-solvability of $\pi(G)$. By the inductive hypothesis, $G/G_{k-1}$ admits maximal subgroups. By the correspondence theorem, $G$ admits maximal subgroups. □

Solvable or nilpotent groups do not always admit maximal subgroups. Can you give an example?

12.17. Definition. A group $G$ satisfies the **maximal condition on subgroups** if for every non-empty subset $\mathcal{S}$ of subgroups contains a maximal element (i.e. a subgroup not contained in any other subgroup of $\mathcal{S}$).

12.18. Exercise. A group satisfies the maximal condition on subgroups if and only if every subgroup of $G$ is finitely generated.

12.19. Exercise. Let $H$ be a subgroup of a group $G$. If $G$ satisfies the maximal condition on subgroups, then so does $H$.

12.20. Exercise. Let $G$ be a group and $N$ be a normal subgroup of $G$. If $G/N$ and $N$ satisfy the maximal condition on subgroups, then so does $G$.

12.21. Proposition. *Super-solvable groups satisfy the maximal condition on subgroups. In particular, every super-solvable group is finitely generated.*

Proof. We proceed by induction on the length of the super-solvable sequence. If the length is one, the result holds as the group is cyclic. So assume the result holds for super-solvable groups with super-solvable series of length $\leq n-1$. Let $G$ be a non-trivial super-solvable group and

$$G = G_0 \supsetneq G_1 \supsetneq \cdots \supsetneq G_n = \{1\}$$

a sequence of normal subgroups of $G$ with cyclic factors. Since $G_1$ is super-solvable (Exercise 12.5), $G_1$ satisfies the maximal condition on subgroups by the inductive hypothesis. By Exercise 12.20, $G$ satisfies the maximal condition on subgroups, as $G/G_1$ is cyclic. □

12.22. Example. The abelian group $\mathbb{Q}$ is nilpotent but not super-solvable, as it is not finitely generated.

If $G$ is a group and $x_1, \ldots, x_{n+1} \in G$, let

$$[x_1, x_2 \ldots, x_{n+1}] = [x_1, [x_2, \ldots, x_{n+1}]], \quad n \geq 1.$$

We will prove in Theorem 12.25 that nilpotent groups are super-solvable if and only if they are finitely generated. For this, we need two lemmas.

12.23. Lemma. *Let $G$ be a finite generated group, say $G = \langle X \rangle$ for some finite set $X$. For $n \geq 2$, let*

$$G_n = \langle g[x_1, \ldots, x_n]g^{-1} : x_1, \ldots, x_n \in X,\, g \in G \rangle.$$

*Then $G_n = \gamma_n(G)$ for all $n \geq 2$.*

Proof. Note that each $G_n$ is normal in $G$. We proceed by induction on $n$. The case $n = 2$ is trivial. So let us assume that $\gamma_{n-1}(G) = G_{n-1}$ for some $n \geq 2$. Let $x_1, \ldots, x_n \in X$. Since $[x_1, \ldots, x_n] \in \gamma_n(G)$, $G_{n-1} \subseteq \gamma_n(G)$. Let $N = G_n$ and $\pi \colon G \to G/N$ be the canonical map. The group $G/N$ is finitely generated. Since

$$[\pi(x_1), [\pi(x_2), \ldots, \pi(x_n)]] = \pi([x_1, \ldots, x_n]) = 1,$$

we obtain that $\pi([x_2, \ldots, x_n]) \in Z(G/N)$. Hence $\pi(g[x_2, \ldots, x_n]g^{-1}) = 1$ for all $g \in G$. By the inductive hypothesis,

$$\pi(\gamma_{n-1}(G)) = \pi(G_{n-1}) \subseteq Z(G/N).$$

Since

$$\pi(\gamma_n(G)) = \pi([G, \gamma_{n-1}(G)]) = [\pi(G), \pi(\gamma_{n-1}(G))] = \{1\},$$

we conclude that $\gamma_n(G) \subseteq N = G_n$. □

12.24. Lemma. *Let $G$ be a finitely generated group. Then $\gamma_n(G)/\gamma_{n+1}(G)$ is finitely generated.*

Proof. Assume that $G = \langle X \rangle$ for some finite set $X$. Write

$$g[x_1, \dots, x_n]g^{-1} = [g, [x_1, \dots, x_n]][x_1, \dots, x_n].$$

By Lemma 12.23, $[g, [x_1, \dots, x_n]] \in \gamma_{n+1}(G) = G_{n+1}$. Then

$$g[x_1, \dots, x_n]g^{-1} \equiv [x_1, \dots, x_n] \bmod \gamma_{n+1}(G).$$

Hence $\gamma_n(G)/\gamma_{n+1}(G)$ is generated by the finite set

$$\{[x_1, \dots, x_n]\gamma_{n+1}(G) : x_1, \dots, x_n \in X\}.$$ □

12.25. Theorem. *Let $G$ be a nilpotent group. Then $G$ is super-solvable if and only if $G$ is finitely generated.*

Proof. If $G$ is super-solvable, it is then finitely generated by Proposition 12.21.

Now assume that the nilpotent group $G$ is finitely generated. By Lemma 12.24, each quotient $\gamma_n(G)/\gamma_{n+1}(G)$ is finitely generated, say by the elements $y_1, \dots, y_m$. Let $\pi\colon G \to G/\gamma_{n+1}(G)$ the canonical map. For $j \in \{1, \dots, m\}$, let

$$K_j = \langle \gamma_{n+1}(G), y_1, \dots, y_j \rangle.$$

Since $[G, K_j] \subseteq [G, \gamma_n(G)] = \gamma_{n+1}(G)$, we obtain that $\pi(K_j)$ is central in $\pi(G)$. Thus $\pi(K_j)$ is normal in $\pi(G)$. Hence $K_j$ is normal in $G$. Each quotient $K_j/K_{j-1}$ is cyclic and generated by $y_jK_{j-1}$. Therefore, in between $\gamma_n(G)$ and $\gamma_{n+1}(G)$, we have constructed a sequence of normal subgroups of $G$ with cyclic factors. Since $G$ is nilpotent, there exists an integer $c$ such that $\gamma_{c+1}(G) = \{1\}$. Hence $G$ is super-solvable. □

12.26. Corollary. *Every finitely generated nilpotent group satisfies the maximal condition on subgroups.*

Proof. This is an immediate consequence of Proposition 12.21 and Theorem 12.25. □

12.27. Theorem. *Let $G$ be a nilpotent finitely generated group. Then $T(G)$ is finite.*

Proof. Since $G$ is nilpotent, $G$ satisfies the maximal condition on subgroups (Corollary 12.26). Thus every subgroup of $G$ is finitely generated. Since $T(G)$ is a subgroup (Theorem 7.10), it is a torsion finitely generated group. Hence $T(G)$ is finite by Theorem 7.14. □

## 13. Huppert's super-solvable theorem

13.1. Theorem (Huppert). *Let $G$ be a finite group such that all its maximal subgroups are of prime index. Then $G$ is super-solvable.*

Proof. See [6, Theorem 10.5.8]. □

## 14. Formanek's zero divisors theorem

We start recalling a conjecture formulated by Kaplansky as [10, Problem 6] in 1957.

14.1. Conjecture (Kaplansky). Let $K$ be a field and $G$ be a torsion-free group. Then $K[G]$ has no zero divisors.

In 1973 Formanek proved the following result:

14.2. Theorem (Formanek). *Let $G$ be a torsion-free super-solvable group and $K$ be a field. Then $K[G]$ has no zero divisors.*

Proof. See [11, Theorem 13.3.9]. □

Conjecture 14.1 is also known to hold, for example, when $G$ admits a bi-order (proved by Malcev and independently by Neumann), when $G$ is polycyclic-by-finite (proved by Brown and Farkas–Snider), or when $G$ has the unique product property (proved by Cohen). In full generality, the conjecture is still open. See [11, Chapter 13] for more information.

## 15. The Schur–Zassenhaus theorem

Recall that a group $Q$ **acts by automorphisms** on a group $K$ if there exists a map $Q \times K \to K$, $(q, k) \mapsto q \cdot k$, such that

**1)** $1 \cdot a = a$ for all $a \in K$,
**2)** $x \cdot (y \cdot a) = (xy) \cdot a$ for all $x, y \in Q$ and $a \in K$,
**3)** $x \cdot 1 = 1$ for all $x \in Q$, and
**4)** $x \cdot (ab) = (x \cdot a)(x \cdot b)$ for all $x \in Q$ and $a, b \in K$,

For example, if $K$ is a normal subgroup of $G$, then $G$ acts by automorphisms on $K$ by conjugation.

15.1. Definition. Let $Q$ and $K$ be groups, where $Q$ acts by automorphisms on $K$. A map $\varphi\colon Q \to K$ is said to be a **1-cocycle** if

$$\varphi(xy) = \varphi(x)(x \cdot \varphi(y))$$

for all $x, y \in Q$.

Let $Q$ and $K$ be groups, where $Q$ acts by automorphisms on $K$. The set of 1-cocycles $Q \to K$ will be denoted by

$$Z^1(Q, K) = \{\delta\colon Q \to K : \delta \text{ is a 1-cocycle}\}.$$

15.2. Example. Let $Q$ be a group acting by automorphisms on $K$. The semidirect product $K \rtimes Q$ is a group $G$ that contains a normal subgroup isomorphic to $K$ and a subgroup isomorphic to such that $G = KQ$ and $K \cap Q = \{1\}$. Under the obvious identifications, $Q$ acts on $K$ by conjugation. For each $k \in K$, the map $Q \to K$, $x \mapsto [k, x] = kxk^{-1}x^{-1}$, is a 1-cocycle.

15.3. Exercise. Let $\varphi\colon Q \to K$ be a 1-cocycle. Prove the following statements:
**1)** $\varphi(1) = 1$.
**2)** $\varphi(y^{-1}) = (y^{-1} \cdot \varphi(y))^{-1} = y^{-1} \cdot \varphi(y)^{-1}$.
**3)** The set $\ker \varphi = \{x \in Q : \varphi(x) = 1\}$ is a subgroup of $Q$.

15.4. Lemma. *Let $G$ be a group with a normal subgroup $N$. If $\varphi\colon G \to N$ is a 1-cocycle (where $G$ acts on $N$ by conjugation) with kernel*

$$K = \ker \varphi = \{g \in G : \varphi(g) = 1\},$$

*then $\varphi(x) = \varphi(y)$ if and only if $xK = yK$. In particular, $(G : K) = |\varphi(G)|$.*

Proof. If $\varphi(x) = \varphi(y)$, then, since

$$\varphi(x^{-1}y) = \varphi(x^{-1})(x^{-1} \cdot \varphi(y)) = \varphi(x^{-1})(x^{-1} \cdot \varphi(x)) = \varphi(x^{-1}x) = \varphi(1) = 1,$$

we obtain that $xK = yK$. Conversely, if $x^{-1}y \in K$, then, since

$$1 = \varphi(x^{-1}y) = \varphi(x^{-1})(x^{-1} \cdot \varphi(y)),$$

we obtain that $\varphi(y) = x \cdot \varphi(x^{-1})^{-1}$. We conclude that $\varphi(x) = \varphi(y)$.

The second claim now is clear, as $\varphi$ is constant in each coclass of $K$ and takes $(G : K)$ different values. □

15.5. Lemma. *Let $G$ be a finite group, $N$ be an abelian normal subgroup of $G$ and $S$, $T$ and $U$ be transversals of $N$ in $G$. Let*

$$d(S,T) = \prod st^{-1} \in N,$$

*where the product runs over all elements $s \in S$ and $t \in T$ such that $sN = tN$. The following statements hold:*

**1)** $d(S,T)d(T,U) = d(S,U)$.
**2)** $d(gS, gT) = gd(S,T)g^{-1}$ *for all* $g \in G$.
**3)** $d(nS, S) = n^{(G:N)}$ *for all* $n \in N$.

Proof. If $s \in S$, $t \in T$ and $u \in U$ are such that $sN = tN = uN$, then, since $N$ is abelian and $(st^{-1})(tu^{-1}) = su^{-1}$, we obtain that

$$d(S,T)d(T,U) = \prod (st^{-1})(tu^{-1}) = \prod su^{-1} = d(S,U).$$

Since $sN = tN$ if and only if $gsN = gtN$ for all $g \in G$,

$$g\left(\prod st^{-1}\right)g^{-1} = \prod gst^{-1}g^{-1} = \prod (gs)(gt)^{-1} = d(gS, gT).$$

Finally, since $N$ is normal in $G$, $nsN = sN$ for all $n \in N$. Thus

$$d(nS, S) = \prod (ns)s^{-1} = n^{(G:N)}. \qquad \square$$

Recall that a subgroup $K$ of $G$ admits a **complement** $Q$ if $G$ factorizes as $G = KQ$ with $K \cap Q = \{1\}$. A typical example is the semidirect product $G = K \rtimes Q$, where $K$ is a normal subgroup of $G$ and $Q$ is a subgroup of $G$ such that $K \cap Q = \{1\}$.

15.6. Exercise. Let $Q$ act by automorphisms on $K$. Prove that there is a bijection between the set of complements of $K$ in $K \rtimes Q$ and the set $Z^1(Q, K)$.

We are now ready to prove the first version of the Schur–Zassenhaus theorem.

15.7. Theorem (Schur–Zassenhaus). *Let $G$ be a finite group and $N$ be an abelian normal subgroup of $G$. If $|N|$ and $(G : N)$ are coprime, then $N$ admits a complement in $G$. Moreover, all complements of $N$ are conjugate.*

Proof. Let $T$ be a transversal of $N$ in $G$ and $\theta\colon G \to N$, $\theta(g) = d(gT, T)$. Since $N$ is abelian, Lemma 15.5 implies that $\theta$ is a 1-cocycle, where $G$ acts on $N$ by conjugation:

$$\begin{aligned}\theta(xy) &= d(xyT, T) = d(xyT, xT)d(xT, T)\\ &= (xd(yT, T)x^{-1})d(xT, T) = (x \cdot \theta(y))\theta(x).\end{aligned}$$

Claim. $\theta|_N\colon N \to N$ is surjective.

If $n \in N$, Lemma 15.5 implies that $\theta(n) = d(nT, T) = n^{(G:N)}$. Since $|N|$ and $(G : N)$ are coprime, there exist $r, s \in \mathbb{Z}$ such that $r|N| + s(G : N) = 1$. Thus

$$n = n^{r|N|+s(G:N)} = (n^s)^{(G:N)} = \theta(n^s).$$

Let $H = \ker\theta$. We prove that $H$ is a complement of $N$. By Exercise 15.3, $H$ is a subgroup of $G$. By Lemma 15.4,

$$|N| = |\theta(G)| = (G : H) = \frac{|G|}{|H|}.$$

Since $N \cap H$ is a subgroup of $N$ and a subgroup of $H$, $N \cap H = \{1\}$, as the numbers $|N|$ and $(G : N) = |H|$ are coprime. Since $|NH| = |N||H| = |G|$, we conclude that $G = NH$. Hence $H$ is a complement of $N$.

We now prove that two complements of $N$ are conjugate. Let $K$ be a complement of $N$ in $G$. Since $NK = G$ and $N \cap K = \{1\}$, $K$ is a transversal of $N$. Let $m = d(T, K) \in N$. Since the restriction map $\theta|_N$ is surjective, there exists $n \in N$ such that $\theta(n) = m$. By Lemma 15.5,

$$kmk^{-1} = kd(T, K)k^{-1} = d(kT, kK) = d(kT, K) = d(kT, T)d(T, K) = \theta(k)m$$

for all $k \in K$. Since $N$ is abelian, $\theta(n^{-1}) = m^{-1}$. Thus

$$\begin{aligned}\theta(nkn^{-1}) &= \theta(n)n\theta(kn^{-1})n^{-1} = m\theta(kn^{-1}) \\ &= m\theta(k)k\theta(n^{-1})k^{-1} = m\theta(k)km^{-1}k^{-1} = 1.\end{aligned}$$

Therefore $nKn^{-1} \subseteq H = \ker\theta$. Since $|K| = (G : N) = |H|$, we conclude that $nKn^{-1} = H$. □

15.8. Theorem (Schur–Zassenhaus). *Let $G$ be a finite group and $N$ be a normal subgroup of $G$. If $|N|$ and $(G : N)$ are coprime, then $N$ admits a complement in $G$.*

Proof. We proceed by induction on $|G|$. If there is a proper subgroup $K$ of $G$ such that $NK = G$, then, since $(K : K \cap N) = (G : N)$ and $|N|$ are coprime, $(K : K \cap N) = (G : N)$ is coprime with $|K \cap N|$. Since $K \cap N$ is normal in $K$, the inductive hypothesis implies that $K \cap N$ admits a complement in $K$. Thus there exists a subgroup $H$ of $K$ such that $|H| = (K : K \cap N) = (G : N)$.

Assume that there is no proper subgroup $K$ of $G$ such that $NK = G$. We may assume that $N \neq \{1\}$ (otherwise, $G$ would be a complement of $N$ in $G$). Since $N$ is contained in every maximal subgroup of $G$ (because, if there is a maximal subgroup $M \subsetneq G$ such that $N \not\subseteq M$, then $NM = G$), it follows that $N \subseteq \Phi(G)$. By Frattini's theorem 10.10, $\Phi(G)$ is nilpotent. Thus $N$ is nilpotent and then $Z(N) \neq \{1\}$. Let $\pi\colon G \to G/Z(N)$ be the canonical map. Since $N$ is normal in $G$ and $Z(N)$ is characteristic in $N$, $Z(N)$ is normal in $G$. Moreover,

$$(\pi(G) : \pi(N)) = \frac{|\pi(G)|}{|\pi(N)|} = \frac{|G/Z(N)|}{|N/N \cap Z(N)|} = (G : N)$$

is coprime with $|N|$. Then $(\pi(G) : \pi(N))$ is coprime with $|\pi(N)|$. By the inductive hypothesis, $\pi(N)$ admits a complement in $G/Z(N)$, say $\pi(K)$ for some subgroup $K$ of $G$. Hence $G = NK$, as $\pi(G) = \pi(N)\pi(K) = \pi(NK)$. Since $K = G$ (because there is no $K$ such that $G = NK$), $\pi(N)$ is abelian, as

$$\pi(Z(N)) = \pi(N) \cap \pi(K) = \pi(N) \cap \pi(G) = \pi(N).$$

Thus $N \subseteq Z(N)$ is abelian. By Theorem 15.7, the subgroup $N$ admits a complement. □

15.9. Theorem (Schur–Zassenhaus conjugation theorem). *Let $G$ be a finite group and $N$ be a normal subgroup of $G$ such that $|N|$ and $(G : N)$ are coprime. If either $N$ or $G/N$ is solvable, then all complements of $N$ in $G$ are conjugate.*

Proof. Let $G$ be a minimal counterexample to the theorem, that is there are complements $K_1$ and $K_2$ of $N$ in $G$ such that $K_1$ and $K_2$ are not conjugate.

Claim. Every subgroup $U$ of $G$ satisfies the assumptions of the theorem with respect to the normal subgroup $U \cap N$.

Since $N$ is normal in $G$, $U \cap N$ is normal in $U$. Moreover, $|U \cap N|$ and $(U : U \cap N)$ are coprime, as $|U \cap N|$ divides $|N|$ and $(U : U \cap N) = (UN : N)$ divides $(G : N)$. If $G/N$ is solvable, then $U/U \cap N$ is solvable, as $U/U \cap N$ is isomorphic to a subgroup of $G/N$. If $N$ is solvable, then so is $U \cap N$.

Claim. If there is a normal subgroup $L$ of $G$ such that $\pi(N)$ is normal in $\pi(G)$, where $\pi\colon G \to G/L$ is the canonical map, then $\pi(G)$ satisfies the theorem's assumptions with respect to $\pi(N)$. In this case, if $H$ is a complement of $N$ in $G$, then $\pi(H)$ is a complement of $\pi(N)$ in $\pi(G)$.

If $N$ is solvable, then so is $\pi(N)$. If $G/N$ is solvable, then so is $\pi(G)/\pi(N) \simeq G/NL$. Moreover, $(\pi(G) : \pi(N)) = \frac{|G/L|}{|N/N\cap L|}$ divides $(G : N)$.

If $H$ is a complement of $N$ in $G$, $|\pi(H)|$ and $|\pi(N)|$ are coprime. Then $\pi(H)$ is a complement of $\pi(N)$, as $\pi(G) = \pi(N)\pi(H) = \pi(NH)$ and $\pi(N) \cap \pi(H) = \{1\}$.

Claim. $N$ is minimal normal in $G$.

Let $M \neq \{1\}$ be a normal subgroup of $G$ such that $M \subseteq N$. Let $\pi\colon G \to G/M$ be the canonical map. Then $\pi(G)$ satisfies the theorem's assumptions with respect to the normal subgroup $\pi(N)$. By the minimality of $|G|$, there exists $x \in G$ such that $\pi(xK_1x^{-1}) = \pi(K_2)$. The subgroup $U = MK_2$ satisfies the theorem's assumptions with respect to the normal subgroup $U \cap N$ of $U$. Since $xK_1x^{-1} \cup K_2 \subseteq U$, we conclude that both $xK_1x^{-1}$ and $K_2$ complement $U \cap N$ in $U$. Hence $MK_2 = G$, as $xK_1x^{-1}$ and $K_2$ are not conjugate and $G$, as $G$ is a minimal counterexample. Therefore $M = N$, as

$$\frac{|K_2|}{|M \cap K_2|} = (MK_2 : M) = (G : M) = \frac{|NK_2|}{|M|} = (N : M)|K_2|.$$

Claim. $N$ is not solvable and $G/N$ is solvable.

Otherwise, by Lemma 1.20, $N$ is abelian (because it is minimal normal). This contradicts Theorem 15.7, as it states that $K_1$ and $K_2$ are conjugate.

Let $p\colon G \to G/N$ be the canonical map and $S$ be a subgroup such that $p(S)$ is minimal normal in $p(G) = G/N$. By Lemma 1.20, $p(S)$ is a $p$-group for some prime number $p$. Since $G = NK_1 = NK_2$ and $N \subseteq S$, Dedekind's lemma 10.5 implies that

$$S = N(S \cap K_1) = N(S \cap K_2).$$

Hence both $S \cap K_1$ and $S \cap K_2$ complement $N$ in $S$. Since $p(S) = p(S \cap K_1) = p(S \cap K_2)$ is a $p$-group, $p$ divides $|S|$. The theorem's assumptions hold for $S$ with respect to the normal subgroup $N$, so $|N|$ and $(S : N)$ are coprime. If $p \mid |N|$, then $p \nmid (S : N) = |S \cap K_1| = |S \cap K_2|$, a contradiction. Thus $p \nmid |N|$ and hence $p \nmid |S|$. This implies that both $S \cap K_1$ and $S \cap K_2$ are Sylow $p$-subgroups of $S$, as

$$|S \cap K_1| = (S : N) = |S \cap K_2|.$$

By Sylow's theorem, there exists $s \in S$ such that

$$S \cap sK_1s^{-1} = S \cap K_2.$$

In particular, $S \neq G$ by the minimality of $G$. Let

$$L = S \cap K_2 = S \cap sK_1s^{-1} \neq \{1\}.$$

Since $S$ is normal in $G$, $sK_1s^{-1} \cup K_2 \subseteq N_G(L)$ (because $L$ is both normal in $sK_1s^{-1}$ and in $K_2$). The subgroups $sK_1s^{-1} \subseteq N_G(L)$ and $K_2 \subseteq N_G(L)$ complement $N \cap N_G(L)$ in $N_G(L)$. Hence $N_G(L) = G$ by the minimality of $G$ (if $N_G(L) \neq G$, then both $sK_1s^{-1}$ and $K_2$ are conjugate in $G$ because they are conjugate in $N_G(L)$). Therefore $L$ is normal in $G$.

Let $\pi_L \colon G \to G/L$ be the canonical map. Since both $\pi_L(K_1)$ and $\pi_L(K_2)$ complement $\pi_L(N)$ in $\pi_L(G)$, the minimality of $|G|$ implies that there exists $g \in G$ such that $\pi_L(gK_1g^{-1}) = \pi_L(K_2)$, that is there exists $g \in G$ such that $(gK_1g^{-1})L = K_2L$. Hence $gK_1g^{-1} \cup K_2 \subseteq \langle K_2, L\rangle = K_2$, because $L \subseteq K_2$. In conclusion, $gK_1g^{-1} = K_2$, a contradiction to the minimality of $|G|$. □

By the Feit–Thompson theorem, in the previous theorem, we do not need to assume that either $N$ or $G/N$ is solvable. Since every group of odd order is solvable and $|N|$ and $(G : N)$ are coprime, one of these groups should have odd order.

15.10. Theorem. *Let $G$ be a finite solvable group and $p$ a prime number dividing $|G|$. There exists a maximal subgroup $M$ of $G$ of index a power of $p$.*

Proof. We proceed by induction on $|G|$. If $G$ is a $p$-group, the result clearly holds. So we may assume that $|G|$ is divisible by at least two different prime numbers. Let $p$ be a prime dividing $|G|$, $N$ be a minimal normal subgroup of $G$ and $\pi\colon G \to G/N$ be the canonical map. Since $G$ is solvable, by Lemma 1.20, $N$ is a $q$-group for some prime $q$. Since $G/N$ is solvable, if $p$ divides $(G : N)$, then, by the inductive hypothesis, $G/N$ has a maximal subgroup $M_1$ of index a power of $p$. By the correspondence theorem, $M = \pi^{-1}(M_1)$ is a maximal subgroup of $G$ of index a power of $p$. $p$. If $p$ does not divide $(G : N)$, then $p$ divides $|N|$. Thus $N \in \mathrm{Syl}_p(G)$. Since $N$ is normal in $G$ and $|N|$ and $|G/N|$ are coprime, by Schur–Zassenhaus theorem 15.8, there exists a complement $K$ of $N$ in $G$, that is $G = NK$ and $N \cap K = \{1\}$. Let $M$ be a maximal subgroup containing $K$. Then $(G : M)$ is a power of $p$. □

We now discuss an application to finite super-solvable groups.

15.11. Definition. A finite group $G$ is said to be **lagrangian** if for each $d$ dividing $|G|$ there exists a subgroup of $G$ of order $d$.

The group $\mathbb{A}_4$ is not lagrangian, as it has no subgroups of order six.

15.12. Theorem. *Every finite super-solvable group is lagrangian.*

Proof. Let $p$ be a prime number dividing $|G|$. Since subgroups of super-solvable groups are super-solvable, it is enough to show that there exists a subgroup of index $p$. Since $G$ is solvable, there exists a maximal subgroup $M$ of index $p^\alpha$ by Theorem 15.10. Since maximal subgroups of super-solvable groups have prime index by Theorem 12.11, we conclude that $\alpha = 1$. □

See [5] for an elementary proof.

## 16. Hall's theory for solvable groups

As an application of the Schur–Zassenhaus theorem, we present Hall's theory of solvable groups. For an elementary presentation, see [8].

16.1. Definition. Let $G$ be a finite group and $\pi$ be a set of prime numbers. We say that $G$ is a **$\pi$-group** if every prime dividing $|G|$ belongs to $\pi$. Similarly, a $\pi$-subgroup of $G$ is a subgroup of $G$ that is also a $\pi$-group.

For a set $\pi$ of prime numbers, we define a $\pi$-number as an integer whose prime divisors belong to $\pi$. The set of prime numbers not belonging to $\pi$ will be denoted as $\pi'$. Thus a $\pi'$-number is an integers not divisible by the prime numbers of $\pi$.

16.2. Definition. Let $G$ be a group and $\pi$ be a set of prime numbers. A subgroup $H$ of $G$ is a **Hall $\pi$-subgroup** if $H$ is a $\pi$-subgroup of $G$ and $(G : H)$ is a $\pi'$-number.

We now prove that a finite solvable group of order $nm$ with $\gcd(n, m) = 1$ always admits a subgroup of order $m$.

16.3. Theorem (Hall's!existence theorem). *Let $\pi$ be a set of prime numbers and $G$ be a finite solvable group. Then $G$ has a Hall $\pi$-subgroup.*

PROOF. Assume that $|G| = nm > 1$ and $\gcd(n, m) = 1$. We want to show that $G$ admits a subgroup of order $m$. We proceed by induction on $|G|$. Let $K$ be a minimal normal subgroup of $G$ and $\pi\colon G \to G/N$ be the canonical map. Since $G$ is solvable, $K$ is a $p$-group (Lemma 1.20).

There are two cases to consider. Assume first that $p$ divides $m$. Since $|G/K| < |G|$, the inductive hypothesis and the correspondence theorem imply that there exists a subgroup $J$ of $G$ containing $K$ such that $\pi(J)$ is a subgroup of $\pi(G) = G/K$ of order $m/|K|$. Then $J$ has order $m$, as

$$m/|K| = |\pi(J)| = \frac{|J|}{|K \cap J|} = (J : K).$$

Assume now that $p$ does not divide $m$. By the inductive hypothesis and the correspondence theorem, there exists a subgroup $H$ of $G$ containing $K$ such that $\pi(H)$ is a subgroup of $G/K$ of order $m$. Since $|H| = m|K|$, $K$ is normal in $H$ and $|K|$ is coprime with $|H : K|$, the Schur–Zassenhaus theorem (Theorem 15.8) implies that there exists a complement $J$ of $K$ in $H$. Hence $J$ is a subgroup of $G$ such that $|J| = m$. □

16.4. EXAMPLE. The group $\mathbb{A}_5$ contains a Hall $\{2, 3\}$-subgroups isomorphic to $\mathbb{A}_4$.

16.5. EXAMPLE. The simple group $\mathbf{PSL}_3(2)$ of order 168 does not contain Hall $\{2, 7\}$-subgroups.

16.6. THEOREM (Hall's conjugation theorem). *Let $G$ be a finite solvable group and $\pi$ be a set of prime numbers. Then all two Hall $\pi$-subgroups of $G$ are conjugate.*

PROOF. We may assume that $G \neq \{1\}$. We proceed by induction on $|G|$. Let $H$ and $K$ be Hall $\pi$-subgroups of $G$. Let $M$ be a minimal normal subgroup of $G$ and $\pi\colon G \to G/M$ be the canonical map. Since $G$ is solvable, $M$ is a $p$-group for some prime number $p$ (Lemma 1.20). Since $\pi(H)$ and $\pi(K)$ are both Hall $\pi$-subgroups of $G/M$, by the inductive hypothesis, the subgroups $\pi(H)$ and $\pi(K)$ are conjugate in $G/M$. Thus there exists $g \in G$ such that $gHMg^{-1} = KM$.

There are two cases to consider. Assume first that $p \in \pi$. Since $|HM|$ and $|KM|$ are $\pi$-numbers and $|H| = |K|$ is the largest $\pi$-number dividing $|G|$, we conclude that $H = HM$ and $K = KM$. In particular, $gHg^{-1} = K$.

Assume now that $p \notin \pi$. Then $K$ admits a complement $M$ in $KM$, as $K \cap M = \{1\}$. We claim that $gHg^{-1}$ complements $M$ in $KM$. Since $M$ is normal in $G$,

$$(gHg^{-1})M = gHMg^{-1} = KM,$$

and $gHg^{-1} \cap M = \{1\}$, as $p \notin \pi$. These complements are conjugate by the Schur–Zassenhaus theorem 15.9. □

16.7. COROLLARY. *Let $G$ be a finite group, $N$ a normal subgroup of $G$ and $n = |N|$. Assume that either $N$ of $G/N$ is solvable. If $|G : N| = m$ is coprime with $n$ and $m_1$ divide a $m$, then every subgroup of $G$ of order $m_1$ is contained in some subgroup of order $m$.*

PROOF. Let $H$ be a complement of $N$ in $G$. Then $|H| = m$. Let $H_1$ be a subgroup of $G$ such that $|H_1| = m_1$. Since $\gcd(n, m) = 1$, $m_1 = |H_1| = |H \cap NH_1|$, as

$$\frac{|H||N||H_1|}{|H \cap NH_1|} = \frac{|H||NH_1|}{|H \cap NH_1|} = |H(NH_1)| = |G| = |NH| = |N||H|.$$

Since both $H_1$ and $H \cap NH_1$ are complements of $N$ in $NH_1$, and both groups have orders coprime with $n$, there exists $g \in G$ such that $H_1 = g(H \cap NH_1)g^{-1}$. Thus $H_1 \subseteq gHg^{-1}$ and hence $|gHg^{-1}| = m$. □

## 17. Subnormality

17.1. Definition. Let $G$ be a group. A subgroup $H$ of $G$ is said to be subnormal in $G$ if there is a sequence of subgroups

$$H = H_0 \subsetneq H_1 \subsetneq \cdots \subsetneq H_k = G$$

with $H_i$ normal in $H_{i+1}$ for all $i \in \{0, \ldots, k-1\}$.

17.2. Example. Let $G = \mathbb{S}_4$. Then $K = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\}$ is normal in $G$. The subgroup $L = \{\mathrm{id}, (12)(34)\}$ is subnormal in $G$ (and not normal).

17.3. Exercise. Prove that the correspondence theorem preserves subnormality.

17.4. Exercise. Let $S$ be a subnormal group of $G$ and $K$ be an arbitrary subgroup of $G$. Prove that $S \cap K$ is subnormal in $K$.

17.5. Theorem. *Let $G$ be a finite group. Then $G$ is nilpotent if and only if every subgroup of $G$ is subnormal in $G$.*

Proof. Assume first that every subgroup of $G$ is subnormal in $G$. Let $H$ be a subnormal subgroup of $G$, where

$$H = H_0 \subsetneq H_1 \subsetneq \cdots \subsetneq H_k = G$$

with $H_i$ normal in $H_{i+1}$. Since $H \subsetneq H_1 \subseteq N_G(H)$, $G$ is nilpotent by Exercise 8.4.

Assume now that $G$ is nilpotent. Let $H$ be a subgroup of $G$. We proceed by induction on $(G : H)$. If $(G : H) = 1$, then $H = G$ and the theorem holds. If $H \neq G$, since $H \subsetneq N_G(H)$ by Lemma 5.18,

$$(G : N_G(H)) < (G : H).$$

By the inductive hypothesis, $N_G(H)$ is subnormal in $G$. Since $H$ is normal in $N_G(H)$, we conclude that $H$ is subnormal in $G$. □

17.6. Corollary. *Let $G$ be a finite group and $K$ be a central subgroup of $G$ (that is, $K \subseteq Z(G)$). Then $G$ is nilpotent if and only if $G/K$ is nilpotent.*

Proof. If $G$ is nilpotent, then so is $G/K$. Conversely, let $\pi\colon G \to G/K$ be the canonical map and $U$ be a subgroup of $G$. Since $G/K$ is nilpotent, Theorem 17.5 implies that $\pi(U)$ is a subnormal subgroup of $G/K$. By the correspondence theorem, $UK$ is a subnormal subgroup of $G$. Since $K$ is central, $U$ is normal in $UK$. Hence $U$ is subnormal in $G$ and therefore $G$ is nilpotent by Theorem 17.5. □

17.7. Theorem. *Let $G$ be a finite group and $H$ be a subgroup of $G$. Then $H$ is nilpotent and subnormal in $G$ if and only if $H \subseteq F(G)$.*

Proof. Assume first that $H \subseteq F(G)$. Since $F(G)$ is nilpotent by Theorem 11.6, so is $H$. Moreover, since $H$ is subnormal in $F(G)$ (Theorem 17.5) and $F(G)$ is normal in $G$, $H$ is subnormal in $G$.

Assume now that $H$ is nilpotent and subnormal in $G$. We proceed by induction on $|G|$. If $H = G$, then the result holds. Assume then that $H \neq G$. Since $H$ is subnormal in $G$, there is a sequence

$$H = H_0 \subsetneq H_1 \subsetneq \cdots \subsetneq H_k = G$$

of subgroups of $G$ with $H_i$ normal in $H_{i+1}$ for all $i$. Let $M = H_{k-1}$. Since $M \neq G$ and $M$ is normal in $G$, $H \subseteq F(M)$ by the inductive hypothesis. Thus $H \subseteq F(M) = M \cap F(G) \subseteq F(G)$ by Corollary 11.9. □

Before proving another important theorem of Wielandt, we need a lemma.

17.8. Lemma. *Let $M$ and $N$ be normal subgroups of $G$ such that $M \cap N = \{1\}$. Then $M \subseteq C_G(N)$.*

Proof. Let $m \in M$ and $n \in N$. Then $[n, m] = (nmn^{-1})m \in M$, since $M$ is normal in $G$ and Moreover, $[n, m] = n(mn^{-1}m^{-1}) \in N$, since $N$ is normal in $G$. Thus $[n, m] \in M \cap N = \{1\}$. □

17.9. Exercise. A group $G$ is said to be **characteristically simple** if $G$ is non-trivial and has no proper characteristic subgroups. Prove that any minimal normal subgroup of $G$ is characteristically simple.

17.10. Definition. Let $G$ be a group. If $G$ admits minimal normal subgroups, the **socle** of $G$ is defined as the subgroup $\mathrm{Soc}(G)$ of $G$ generated by all minimal normal subgroups of $G$. If $G$ admits no minimal normal subgroups, then $\mathrm{Soc}(G) = \{1\}$.

For example, $\mathrm{Soc}(\mathbb{Z}) = \{0\}$ and $\mathrm{Soc}(\mathbf{SL}_2(3)) \simeq C_2$.

17.11. Exercise. Prove that the socle of a group is a characteristic subgroup.

Let $G$ be a finite group and $N$ be a non-trivial normal subgroup of $G$. Then $N$ contains a minimal normal subgroup of $G$. Since $\mathrm{Soc}(G)$ is generated by the minimal normal subgroups of $G$, $N \cap \mathrm{Soc}(G) \neq \{1\}$. In particular, the socle of a non-trivial subgroup is always non-trivial.

17.12. Exercise. Prove that the socle of a group is a direct product of minimal normal subgroups.

17.13. Exercise. Prove the following statements:

**1)** A direct product of isomorphic simple groups is characteristically simple.

**2)** A characteristically simple group with at least one minimal normal subgroup is a direct product of isomorphic simple groups.

17.14. Theorem (Wielandt). *Let $G$ be a finite group. If $S$ is a subnormal group of $G$ and $M$ is a minimal normal subgroup of $G$, then $M \subseteq N_G(S)$.*

Proof. We proceed by induction on $|G|$. If $S = G$ the result holds. So assume that $S \neq G$. Since $S$ is subnormal in $G$, there exists a sequence

$$S = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_{k-1} \subsetneq S_k = G$$

of subgroups of $G$ such that $S_i$ is normal in $S_{i+1}$ for all $i$. Let $N = S_{k-1}$.

If $M \cap N \neq \{1\}$, then $M \subseteq N$ (because since $M$ and $N$ are both normal in $G$, $M \cap N = M$ by the minimality of $M$). We claim that $M \subseteq \mathrm{Soc}(N)$. Since $M \neq \{1\}$ and $M$ is normal in $N$, $M \cap \mathrm{Soc}(N) \neq \{1\}$. Moreover, since $\mathrm{Soc}(N)$ is characteristic in $N$ and $N$ is normal in $G$, it follows that $\mathrm{Soc}(N)$ is normal in $G$. Hence $M \cap \mathrm{Soc}(N)$ is a normal subgroup of $G$. Since $\{1\} \neq M \cap \mathrm{Soc}(N) \subseteq M$, we conclude that $M \cap \mathrm{Soc}(N) = M$ by the minimality of $M$. By the inductive hypothesis, every minimal normal subgroup of $N$ normalizes $S$. Thus $\mathrm{Soc}(N) \subseteq N_N(S) \subseteq N_G(S)$ and therefore

$$M \subseteq \mathrm{Soc}(N) \subseteq N_G(S).$$

If $M \cap N = 1$, Lemma 17.8 implies that

$$M \subseteq C_G(N) \subseteq C_G(S) \subseteq N_G(S). \qquad \square$$

17.15. Corollary. *Let $G$ be a finite group and $S$ be a subnormal subgroup of $G$. Then*

$$\mathrm{Soc}(G) \subseteq N_G(S).$$

Proof. By Theorem 17.14, every minimal normal subgroup of $G$ is contained in $N_G(S)$. Then $\mathrm{Soc}(G) = \langle M : M \text{ minimal normal subgroup of } G \rangle \subseteq N_G(S)$. $\square$

17.16. Theorem (Wielandt). *Let $G$ be a finite group and $S$ and $T$ be subnormal subgroups of $G$. Then $S \cap T$ and $\langle S, T \rangle$ are subnormal in $G$.*

Proof. We first prove that $S \cap T$ is subnormal in $G$. Since subnormality is a transitive relation, it is enough to see that $S \cap T$ is subnormal in $T$. Since $S$ is subnormal in $G$, there exists a sequence

$$S = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_k = G$$

of subgroups of $G$ such that $S_i$ is normal in $S_{i+1}$ for all $i$. Each $S_{j-1} \cap T$ is normal in $S_j \cap T$. Then $S \cap T$ is subnormal in $T$.

We now prove that $\langle S, T \rangle$ is subnormal in $G$ We proceed by induction on $|G|$. Assume that $G \neq \{1\}$. Let $M$ be a minimal normal subgroup of $G$ and $\pi \colon G \to G/M$ be the canonical map. Since both $\pi(S)$ and $\pi(T)$ subnormal in $G/M$ and $|G/M| < |G|$, the inductive hypothesis implies that

$$\pi(\langle S, T \rangle M) = \pi(\langle S, T \rangle) = \langle \pi(S), \pi(T) \rangle$$

is subnormal in $G/M$. By the correspondence theorem, $\langle S, T \rangle M$ is subnormal in $G$. Theorem 17.14 implies that $M \subseteq N_G(S)$ and $M \subseteq N_G(T)$. Hence $M \subseteq N_G(\langle S, T \rangle)$. Since $\langle S, T \rangle$ is normal in $\langle S, T \rangle M$ and $\langle S, T \rangle M$ is subnormal in $G$, we conclude that $\langle S, T \rangle$ is subnormal in $G$. $\square$

## 18. Wielandt's zipper theorem

18.1. Theorem (Wielandt). *Let $G$ be a finite group and $S$ be a subgroup of $G$ subnormal in every proper subgroup of $G$ containing $S$. If $S$ is not subnormal in $G$, then there exists a unique maximal subgroup of $G$ containing $S$.*

Proof. We proceed by induction on $(G : S)$. If $S$ is not subnormal in $G$, then $S \neq G$ and the case where $(G : S) = 1$ holds.

Since $S$ is not subnormal in $G$, $N_G(S) \neq G$. Then $S \subseteq N_G(S) \subseteq M$ for some maximal subgroup $M$ of $G$. Assume that $S \subseteq K$ for some maximal subgroup $K$ of $G$. We claim that que $K = M$. Since $S \subseteq K \neq G$, $S$ is subnormal in $K$ by assumption. If $S$ is normal in $K$,

then $K \subseteq N_G(S) \subseteq M$. Hence $K = M$, by the maximality of $K$. If $S$ is not normal in $K$, there exist some shortest sequence $S_0, \ldots, S_m$ of subgroups of $K$ such that

$$S = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_m = K,$$

where $S_i$ is normal in $S_{i+1}$ for all $i$. By the minimality of $m$, $S$ is not normal in $S_2$. Let $x \in S_2$ be such that $xSx^{-1} \neq S$ and $T = \langle S, xSx^{-1} \rangle \subseteq K$.

Note that $xSx^{-1}$ satisfies the assumptions on $S$. In fact, let $H$ be a proper subgroup of $G$. Then $S$ is subnormal in $x^{-1}Hx$ and therefore $xSx^{-1}$ is subnormal in $H$.

Since $xSx^{-1} \subseteq xS_1x^{-1} = S_1 \subseteq N_G(S)$, we obtain that $T \subseteq N_G(S) \subseteq M$. Moreover, $S$ is normal in $T$. Thus $T \neq G$.

We claim that $T$ satisfies the assumptions on $S$. If $T$ is subnormal in $G$, then, since $S$ is normal in $T$, $S$ is subnormal in $G$. If $H$ is a proper subgroup of $G$ containing $T$, then, since $S \subseteq H$, $S$ is subnormal in $H$. Moreover, $xSx^{-1}$ is subnormal in $H$. Hence $T$ is subnormal in $H$ by Theorem 17.16.

Since $S \subsetneq T$, $(G : T) < (G : S)$. By the inductive hypothesis, $T$ is contained in a unique maximal subgroup of $G$. Therefore $K = M$, since $T \subseteq M$ and $T \subseteq K$. □

Before giving an application, we need a lemma.

18.2. Lemma. *Let $G$ be a group and $H$ be a subgroup of $G$. If $(xHx^{-1})H = G$ for some $x \in G$, then $H = G$.*

Proof. Write $x = uv$ for some $u \in xHx^{-1}$ and $v \in H$. Since $u \in xHx^{-1}$ and $u^{-1}x = v \in H$, we obtain that $H = vHv^{-1} = u^{-1}(xHx^{-1})u = xHx^{-1}$. Thus $G = H$. □

Recall that two subgroups $S$ and $T$ of a group $G$ are said to be **permutable** if $ST = TS$.

18.3. Theorem. *Let $G$ be a finite group and $S$ be a subgroup of $G$ permutable with any of its conjugates. Then $S$ is subnormal in $G$.*

Proof. We proceed by induction on $|G|$. Assume that $S$ is subnormal in every subgroup $H$ such that $S \subseteq H \subsetneq G$. If $S$ is not subnormal in $G$, then, by Theorem 18.1, there exists a unique maximal subgroup $M$ of $G$ such that $S \subseteq M$. Let $x \in G$ and $T = xSx^{-1}$. By Lemma 18.2, $ST \neq G$ (because $S \neq G$). Thus $ST$ is contained in some maximal subgroup of $G$. Since $S \subseteq ST$ and $S$ is contained in a unique maximal subgroup of $G$, we conclude that $T \subseteq ST \subseteq M$. Since $S^G = \langle xSx^{-1} : x \in G \rangle \subseteq M \neq G$, the inductive hypothesis implies that $S$ is subnormal in $S^G$. Hence $S$ is subnormal in $G$ since $S^G$ is normal in $G$, a contradiction. □

## 19. Baer's theorem

19.1. Theorem (Baer). *Let $G$ be a finite group and $H$ be a subgroup of $G$. Then $H \subseteq F(G)$ if and only if $\langle H, xHx^{-1}\rangle$ is nilpotent for all $x \in G$.*

Proof. If $H \subseteq F(G)$, then $xHx^{-1} \subseteq F(G)$ for all $x \in G$, since $F(G)$ is normal in $G$. Thus $\langle H, xHx^{-1}\rangle$ is nilpotent, as it is a subgroup of $F(G)$.

Conversely, assume that $\langle H, xHx^{-1}\rangle$ is nilpotent for all $x \in G$. Since $H \subseteq \langle H, xHx^{-1}\rangle$, $H$ is nilpotent. By Theorem 17.7, it is enough to see that $H$ is subnormal in $G$. Suppose that $H$ is not subnormal in $G$. In particular, $H$ is properly contained in some proper subgroup $K$ of $G$. Since $\langle H, kHk^{-1}\rangle$ is nilpotent for all $k \in K$, $H$ is subnormal in $K$ if we assume that $G$ is a minimal counterexample. By Theorem 18.1, there exists a unique maximal subgroup $M$ of $G$ containing $H$.

Assume first that $G = \langle H, xHx^{-1}\rangle$ for some $x \in G$. Since $G$ is nilpotent, $H$ subnormal in $G$ by Theorem 17.5, a contradiction.

Assume now that $\langle H, xHx^{-1}\rangle \neq G$ for all $x \in G$. For each $x \in G$, there exists a maximal subgroup containing $\langle H, xHx^{-1}\rangle$. Since $H \subseteq \langle H, xHx^{-1}\rangle$ and $H$ is contained in a unique maximal subgroup, we conclude that $\langle H, xHx^{-1}\rangle \subseteq M$ for all $x \in G$. In particular, the normal closure $H^G$ of $H$ is properly contained in $G$. By the inductive hypothesis, $H$ is subnormal in $H^G$ and $H^G$ is normal in $G$, we conclude that $H$ is subnormal in $G$, a contradiction. □

## 20. Zenkov's theorem

20.1. Theorem (Zenkov). *Let $G$ be a finite group and $A$ and $B$ be abelian subgroups of $G$. Let $M \in \{A \cap gBg^{-1} : g \in G\}$ such that no $A \cap gBg^{-1}$ is properly contained in $M$. Then $M \subseteq F(G)$.*

Proof. Suppose the result is not true and let $G$ be a minimal counterexample. Without loss of generality, we may assume that $M = A \cap B$.

Assume first that $G = \langle A, gBg^{-1}\rangle$ for some $g \in G$. Since $A$ and $B$ are both abelian,

$$A \cap gBg^{-1} \subseteq Z(G)$$

and hence

$$A \cap gBg^{-1} = g^{-1}(A \cap gBg^{-1})g \subseteq A \cap B = M.$$

By the minimality of $G$,

$$M = A \cap gBg^{-1} \subseteq Z(G) \subseteq F(G)$$

by Corollary 11.7.

Assume now that $G \neq \langle A, gBg^{-1}\rangle$ for all $g \in G$. Let $g \in G$, $H = \langle A, gBg^{-1}\rangle \neq G$ and $C = B \cap H$. Since $A \subseteq H$, we obtain that $M = A \cap B = A \cap C$ and $A \cap hCh^{-1} = A \cap hBh^{-1}$ for all $h \in H$. This implies that no $A \cap hCh^{-1}$ is properly contained in $A \cap C$. By the inductive hypothesis on $H$,

$$M = A \cap B = A \cap C \subseteq F(H).$$

We now prove that every Sylow $p$-subgroup $P$ of $M$ is contained in $F(G)$. Moreover, $M \subseteq F(G)$. If $P \in \mathrm{Syl}_p(M)$, then $P \subseteq M \subseteq F(H)$. Since $O_p(H)$ is the only Sylow $p$-subgroup of $F(H)$, $P \subseteq O_p(H)$. Since $P \subseteq M \subseteq B$,

$$gPg^{-1} \subseteq gBg^{-1} \subseteq H.$$

Thus $O_p(H)(gPg^{-1})$ is a $p$-subgroup of $H$ containing $\langle P, gPg^{-1}\rangle$. Hence $\langle P, gPg^{-1}\rangle$ is nilpotent for all $g \in G$, since it is a $p$-group. By Baer's theorem 19.1, $P \subseteq F(G)$ for all Sylow $p$-subgroup $P$ of $M$. □

20.2. Corollary. *Let $G$ be a non-trivial finite group and $A$ be an abelian subgroup of $G$ such that $|A| \geq (G : A)$. Then $A \cap F(G) \neq \{1\}$.*

Proof. Let $g \in G$. We may assume that $G \neq A$. Then $(gAg^{-1})A \neq G$ by Lemma 18.2. Since $|gAg^{-1}||A| = |A|^2 \geq |A|(G : A) = |G|$,

$$|G| > |gAg^{-1}A| = \frac{|A||gAg^{-1}|}{|A \cap gAg^{-1}|} \geq \frac{|G|}{|A \cap gAg^{-1}|}.$$

Hence $A \cap gAg^{-1} \neq \{1\}$ for all $g \in G$. By Zenkov's theorem, the trivial subgroup is not contained in the set $\{A \cap gAg^{-1} : g \in G\}$. A minimal subset of this set then belongs to $F(G)$. □

20.3. Corollary. *Let $G = NA$ be a finite group, where $N$ is a normal subgroup of $G$, $A$ is an abelian subgroup of $G$ and $C_A(N) = \{1\}$. If $F(N) = \{1\}$, then $|A| < |N|$.*

Proof. Since $N$ is normal in $G$,

$$N \cap F(G) = F(N) = \{1\}$$

by Corollary 11.9. Thus $[N, F(G)] = \{1\}$, since both $N$ and $F(G)$ are normal in $G$. Since

$$|A| \geq |N| \geq \frac{|N|}{|N \cap A|} = (NA : A) = (G : A),$$

$A \cap F(G) \neq \{1\}$ by Corollary 20.2. If $1 \neq a \in A \cap F(G)$, then $a \in C_A(N) = \{1\}$, a contradiction. □

## 21. Brodkey's theorem

21.1. Theorem (Brodkey). *Let $G$ be a finite group such that there exists an abelian $P \in \mathrm{Syl}_p(G)$. Then there exist $S, T \in \mathrm{Syl}_p(G)$ such that $S \cap T = O_p(G)$.*

Proof. By applying Zenkov's theorem (Theorem 20.1) with $A = B = P$,

$$P \cap gPg^{-1} \subseteq F(G)$$

for some $g \in G$. Since $O_p(G)$ is the only Sylow $p$-subgroup of $F(G)$, $P \cap gPg^{-1} \subseteq O_p(G)$. Hence $P \cap gPg^{-1} = O_p(G)$, since $O_p(G)$ is contained in every Sylow $p$-subgroup of $G$. □

21.2. Corollary. *Let $G$ be a finite group. If there exists an abelian $P \in \mathrm{Syl}_p(G)$,*

$$(G : O_p(G)) \leq (G : P)^2.$$

Proof. By Brodkey's theorem, there exist $S, T \in \mathrm{Syl}_p(G)$ such that $S \cap T = O_p(G)$. Then

$$|G| \geq |ST| = \frac{|S||T|}{|S \cap T|} = \frac{|P|^2}{|O_p(G)|},$$

which implies the claim. □

21.3. Corollary. *Let $G$ be a finite group. If there exists an abelian $P \in \mathrm{Syl}_p(G)$ such that $|P| > \sqrt{|G|}$, then $O_p(G) \neq \{1\}$.*

Proof. Since $(G : P)^2 < |G|$, the previous corollary implies that $O_p(G) \neq \{1\}$. □

## 22. Lucchini's theorem

22.1. Theorem (Lucchini). *Let $G$ be a finite group and $A$ be a proper cyclic subgroup of $G$. If $K = \mathrm{Core}_G(A)$, then $(A : K) < (G : A)$.*

Proof. We proceed by induction on $|G|$. Let $\pi\colon G \to G/K$ be the canonical map. Note that $\mathrm{Core}_{G/K}\,\pi(A)$ is trivial.

Assume first that $K \neq \{1\}$. Since $\pi(A)$ is a proper cyclic subgroup of $G/K$ and $K \subseteq A$, the inductive hypothesis implies that

$$(A : K) = |\pi(A)| = (\pi(A) : \pi(K)) < (\pi(G) : \pi(A)) = \frac{(G : K)}{(A : K)} = (G : A).$$

Assume now that $K = \{1\}$. We want to prove that $|A| < (G : A)$. Suppose that $|A| \geq (G : A)$. Since $A \neq G$, $A \cap F(G) \neq \{1\}$ by Corollary 20.2. In particular, $F(G) \neq \{1\}$. Let $E$ be a minimal normal subgroup of such that $E \subseteq F(G)$. By Theorem 7.1, $E \cap Z(F(G)) \neq \{1\}$. Since $E \cap Z(F(G))$ is normal in $G$ and $E$ is minimal, $E \cap Z(F(G)) = E$, that is $E \subseteq Z(F(G))$. In particular, $E$ is abelian. By the minimality of $E$, there is a prime number $p$ such that $x^p = 1$ for all $x \in E$.

Claim. $A \cap F(G)$ is a normal subgroup of $EA$.

Since $E$ is normal in $G$, $EA$ is a subgroup of $G$. Since $A \cap F(G) \subseteq A$, $A \cap F(G)$ is a subgroup of $EA$. Since $F(G)$ is normal in $G$, $a(A \cap F(G))a^{-1} = A \cap F(G)$ for all $a \in A$. Moreover, $E \subseteq Z(F(G))$ and $A \cap F(G) \subseteq F(G)$ imply that $x(A \cap F(G))x^{-1} = A \cap F(G)$ for all $x \in E$.

Claim. $EA \neq G$.

If $G = EA$, then, since $A \cap F(G)$ is a normal subgroup of $G$ contained in $A$, we conclude that $\{1\} \neq A \cap F(G) \subseteq K = \{1\}$, a contradiction.

Let $p\colon G \to G/E$ the canonical map. By the correspondence theorem, there exists a normal subgroup $M$ of $G$ such that $E \subseteq M$ and $p(M) = \mathrm{Core}_{G/E}(p(A))$. Since $EA \neq G$, $p(A)$ is a proper cyclic subgroup of $p(G)$. Since $p(A) \simeq A/A \cap E \simeq EA/E$ and $p(M) \simeq M/E$, the inductive hypothesis implies that $(EA : M) < (G : EA)$, as

$$\frac{|EA/E|}{|M/E|} = (p(A) : p(M)) < (p(G) : p(A)) = \frac{|G/E|}{|EA/E|}.$$

Claim. $MA = EA$.

Since $E \subseteq M$, $EA \subseteq MA$. Conversely, if $m \in M$, then, since $p(m) \in \mathrm{Core}_{G/E}(p(A))$, we obtain that $p(m) \in p(A)$. Thus $m \in EA$.

Let $B = A \cap M$. Since $(AE : M) < (G : EA)$,

$$(A : B) = |A/A \cap M| = |AM/M| = (EA : M).$$

By the inductive hypothesis,

$$\begin{aligned}(M : B) &= (M : A \cap M) = (MA : A)\\ &= (EA : A) = \frac{(G : A)}{(G : EA)} < \frac{(G : A)}{(AE : M)} = \frac{(G : A)}{(A : B)} \leq |B|,\end{aligned} \tag{22.1}$$

as $|A| \geq (G : A)$.

Claim. $M = EB$.

Since $E \cup B \subseteq M$, $EB \subseteq M$. Conversely, if $m \in M$, then $m = ea$ for some $e \in E$ and $a \in A$. Since $e^{-1}m = a \in A \cap M = B$ (because $E \subseteq M$), $m \in EB$.

Claim. $M$ is not abelian.

Suppose that $M$ is abelian. The map $f\colon M \to M$, $m \mapsto m^p$, is a group homomorphism such that $E \subseteq \ker f$. Since $M = EB$, $f(M) \subseteq f(B) \subseteq B \subseteq A$. Since $M$ is normal in $G$, $f(M)$ is normal in $G$. Thus $f(M) = \{1\}$, as $f(M) \subseteq \mathrm{Core}_G(A) = \{1\}$ is the largest normal subgroup of $G$ contained in $A$. In particular, since $B$ is normal in $M = EB$ (because $M$ is normal), $M/B$ is a $p$-group. Since $B \subseteq M$, $f(B) = \{1\}$. Moreover, since $B \subseteq A$ is cyclic, (22.1) implies that $(M : B) < |B| \leq p$. Thus $M = B \subseteq A$ and $M = E = \{1\}$ (because $M$ is normal in $G$ and $\mathrm{Core}_G(A) = K = \{1\}$ is the largest normal subgroup of $G$ containing $A$), a contradiction.

Claim. $Z(M)$ is cyclic.

Since $M$ is not abelian and $M/E = EB/E \simeq B/E \cap B$ is cyclic, $E \not\subseteq Z(M)$, that is $E \cap Z(M) \subsetneq E$. Thus

$$E \cap Z(M) = \{1\} \tag{22.2}$$

by the minimality of $E$. Hence

$$Z(M) = Z(M)/Z(M) \cap E \simeq p(Z(M)) \subseteq p(M) = \mathrm{Core}_{G/E}\, p(A) \subseteq p(A)$$

and therefore $Z(M)$ is cyclic, since $p(A)$ is cylic.

Since $B \subseteq A$ is abelian and $(M : B) < |B|$, $B \cap F(M) \neq \{1\}$ by Corollary 20.2. Then

$$[E, F(M)] = \{1\},$$

because $E \subseteq Z(F(G))$ and $F(M) \subseteq F(G)$ by Corollary 11.9). Hence $B \cap F(M) \subseteq Z(M)$, since $M = BE$, $[B \cap F(M), E] \subseteq [F(M), E] = \{1\}$ and $[B \cap F(M), B] = 1$ as $B$ is abelian. Since $Z(M)$ is cyclic, $B \cap F(M)$ is characteristic in $Z(M)$. Since $Z(M)$ is normal in $G$, $\{1\} \neq B \cap F(M)$ is a normal subgroup of $G$ contained in $A$, a contradiction. □

## 23. Horosevskii's theorem

To conclude this section, we present a striking application of Lucchini's theorem.

23.1. Corollary (Horosevskii). *Let $G$ be a finite non-trivial group and $\sigma \in \mathrm{Aut}(G)$. Then $|\sigma| < |G|$.*

Proof. Let $A = \langle \sigma \rangle$ act by automorphisms on $G$ and $\Gamma = G \rtimes A$. The group operation of $\Gamma$ is

$$(g, \sigma^k)(h, \sigma^l) = (g\sigma^k(h), \sigma^{k+l}).$$

Identify $A$ with $\{1\} \times A$ and $G$ with $G \times \{1\}$. Let $K = \mathrm{Core}_\Gamma(A) \subseteq A$. Since both $K$ and $G$ are normal in $\Gamma$,

$$[K, G] \subseteq K \cap G \subseteq A \cap G = \{1\}.$$

Thus $K \subseteq A \cap C_\Gamma(G) = \{1\}$. By Lucchini's theorem, $(A : K) < (\Gamma : A)$, that is

$$|\sigma| = |A| = (A : K) < (\Gamma : A) = |G|.$$ □

## 24. The transfer map

If $H$ is a subgroup of $G$, recall that a **transversal** of $H$ in $G$ is a complete set of coset representatives of $G/H$.

24.1. Lemma. *Let $G$ be a group and $H$ be a subgroup of $G$ of finite index. Let $R$ and $S$ be transversals of $H$ in $G$ and let $\alpha\colon H \to H/[H,H]$ be the canonical map. Then*

$$d(R,S) = \prod \alpha(rs^{-1}),$$

*where the product is taken over all pairs $(r,s) \in R \times S$ such that $Hr = Hs$, is well-defined and satisfies the following properties:*

**1)** $d(R,S)^{-1} = d(S,R)$.
**2)** $d(R,S)d(S,T) = d(R,T)$ *for all transversal $T$ of $H$ in $G$.*
**3)** $d(Rg,Sg) = d(R,S)$ *for all $g \in G$.*
**4)** $d(Rg,R) = d(Sg,S)$ *for all $g \in G$.*

Proof. The product that defines $d(R,S)$ is well-defined since $H/[H,H]$ is an abelian group. The first three claim are trivial. Let us prove 4). By 2),

$$d(Rg,Sg)d(Sg,S)d(S,R) = d(Rg,S)d(S,R) = d(Rg,R).$$

Since $H/[H,H]$ is abelian, 1) and 3) imply that

$$d(Rg,Sg)d(Sg,S)d(S,R) = d(R,S)d(S,R)d(Sg,S) = d(Sg,S). \qquad \square$$

24.2. Theorem. *Let $G$ be a group and $H$ be a finite-index subgroup of $G$. The map*

$$\nu\colon G \to H/[H,H], \quad g \mapsto d(Rg,R),$$

*does not depend on the transversal $R$ of $H$ in $G$ and is a group homomorphism.*

Proof. The previous lemma implies that the map does not depend on the transversal used. Moreover, $\nu$ is a group homomorphism, as

$$\nu(gh) = d(R(gh),R) = d(R(gh),Rh)d(Rh,R) = d(Rg,R)d(Rh,R) = \nu(g)\nu(h). \qquad \square$$

The theorem justifies the following definition:

24.3. Definition. Let $G$ be a group and $H$ be a finite-index subgroup of $G$. The **transfer map** of $G$ in $H$ is the group homomorphism

$$\nu\colon G \to H/[H,H], \quad g \mapsto d(Rg,R),$$

of Theorem 24.2, where $R$ is some transversal of $H$ in $G$.

We need methods for computing the transfer map. If $H$ is a subgroup of $G$ and $(G : H) = n$, let $T = \{x_1, \ldots, x_n\}$ be a transversal of $H$. For $g \in G$ let

$$\nu(g) = \prod \alpha(xy^{-1}),$$

where the product is taken over all pairs $(x,y) \in (Tg) \times T$ such that $Hx = Hy$ and $\alpha\colon H \to H/[H,H]$ is the canonical map. If we write $x = x_i g$ for some $i \in \{1, \ldots, n\}$, then $Hx_i g = Hx_{\sigma(i)}$ for some permutation $\sigma \in \mathbb{S}_n$. Thus

$$\nu(g) = \prod_{i=1}^{n} \alpha(x_i g x_{\sigma(i)}^{-1}).$$

The cycle structure of $\sigma$ turns out to be important. For example, if $\sigma = (12)(345)$ and $n = 5$, then a direct calculation shows that

$$\prod_{i=1}^{5} \alpha\left(x_i g x_{\sigma(i)}^{-1}\right) = \alpha(x_1 g^2 x_1^{-1})\alpha(x_3 g^3 x_3^{-1}).$$

This is precisely the content of the following lemma.

24.4. Lemma. *Let $G$ be a group and $H$ be a subgroup of index $n$. Let $T = \{t_1, \ldots, t_n\}$ be a transversal of $H$ in $G$. For each $g \in G$ there exist $m \in \mathbb{Z}_{>0}$ and elements $s_1, \ldots, s_m \in T$ and positive integers $n_1, \ldots, n_m$ such that $s_i^{-1} g^{n_i} s_i \in H$, $n_1 + \cdots + n_m = n$ and*

$$\nu(g) = \prod_{i=1}^{m} \alpha(s_i^{-1} g^{n_i} s_i).$$

Proof. For each $i$ there exist $h_1, \ldots, h_n \in H$ and $\sigma \in \mathbb{S}_n$ such that $g t_i = t_{\sigma(i)} h_i$. Write $\sigma$ as a product of disjoint cycles, say

$$\sigma = \alpha_1 \cdots \alpha_m.$$

Let $i \in \{1, \ldots, n\}$ and write $\alpha_i = (j_1 \cdots j_{n_i})$. Since

$$g t_{j_k} = t_{\sigma(j_k)} h_{j_k} = \begin{cases} t_{j_1} h_{j_k} & \text{if } k = n_i, \\ t_{j_{k+1}} h_{j_k} & \text{otherwise,} \end{cases}$$

then

$$\begin{aligned} t_{j_1}^{-1} g^{n_i} t_{j_1} &= t_{j_1}^{-1} g^{n_i - 1} g t_{j_1} \\ &= t_{j_1}^{-1} g^{n_i - 1} t_{j_2} h_{j_1} \\ &= t_{j_1}^{-1} g^{n_i - 2} g t_{j_2} h_{j_1} \\ &= t_{j_1}^{-1} g^{n_i - 2} t_{j_3} h_{j_2} h_{j_1} \\ &\;\;\vdots \\ &= t_{j_1}^{-1} g t_{j_{n_i}} h_{n_{i-1}} \cdots h_{j_2} h_{j_1} \\ &= t_{j_1}^{-1} t_{j_1} h_{j_{n_i}} \cdots h_{j_2} h_{j_1} \in H. \end{aligned}$$

Thus $s_i = t_{j_1} \in T$. It only remains to note that $\nu(g) = h_1 \cdots h_n$. □

Gauss's lemma in number theory gives conditions for an integer to be a quadratic residue. The lemma appears in some proof of the quadratic reciprocity. Gauss's Lemma is basically a computation of the transfer homomorphism.

24.5. Exercise (Gauss' lemma). Let $p$ be a prime number. Let $G = \mathbb{F}_p^\times$ and $H = \{-1, 1\}$.

**1)** Prove that the transfer homomorphism

$$\nu\colon G \to H, \quad \nu(x) = x^{\frac{p-1}{2}} = \left(\frac{x}{p}\right) = \begin{cases} 1 & \text{if } x \text{ is a square,} \\ -1 & \text{otherwise.} \end{cases}.$$

**2)** For a transversal $T = \{1, 2, \ldots, \frac{p-1}{2}\}$ and elements $x \in G$ and $t \in T$, let

$$\epsilon(x,t) = \begin{cases} 1 & \text{if } xt \in T, \\ -1 & \text{if } xt \notin T. \end{cases}$$

Prove that

$$\left(\frac{x}{p}\right) = \prod_{t \in T} \epsilon(x,t).$$

## 25. Other applications of the transfer homomorphism

25.1. Lemma. *Let $G$ be a group, $H$ be a finite-index subgroup and $n = (G : H)$. Let $S = \{s_1, \ldots, s_n\}$ and $T = \{t_1, \ldots, t_n\}$ be transversals of $H$ in $G$. For each $g \in G$, there exist unique $h_1, \ldots, h_n \in H$ and $\sigma \in \mathbb{S}_n$ such that*

$$gt_i = s_{\sigma(i)} h_i, \quad i \in \{1, \ldots, n\}.$$

Proof. If $i \in \{1, \ldots, n\}$, there exists a unique $j \in \{1, \ldots, n\}$ such that $gt_i \in s_j H$. Thus $gt_i = s_j h_i$ for a unique $h_i \in H$. Thus we have constructed a $\sigma\colon \{1, \ldots, n\} \to \{1, \ldots, n\}$, $\sigma(i) = j$. We need to show that $\sigma \in \mathbb{S}_n$. It is enough to prove that $\sigma$ is injective. If $\sigma(i) = \sigma(k) = j$, since $gt_i = s_j h_i$ and $gt_k = s_j h_k$, we obtain that $t_i^{-1} t_k = h_i^{-1} h_k \in H$. Hence $i = k$, since $t_i H = t_k H$. □

25.2. Theorem. *Let $G$ be a finite group and $p$ be a prime number dividing $|[G,G] \cap Z(G)|$. If $P \in \mathrm{Syl}_p(G)$, then $P$ is non-abelian.*

Proof. Assume that $P$ is abelian. Let $T = \{t_1, \ldots, t_n\}$ be a transversal of $P$ in $G$. Since $[G,G] \cap Z(G)$ is a normal subgroup of $G$, we may assume that $P \cap [G,G] \cap Z(G) \neq \{1\}$. Let $z \in P \cap [G,G] \cap Z(G) \setminus \{1\}$.

Let $\nu\colon G \to P$ be the transfer homomorphism. We compute $\nu(z)$ with Lemma 25.1. For $i \in \{1, \ldots, n\}$, let $x_1, \ldots, x_n \in P$ and $\sigma \in \mathbb{S}_n$ be such that $zt_i = t_{\sigma(i)} x_i$. Since $z \in Z(G)$, $t_i = t_{\sigma(i)} x_i z^{-1}$. By the uniqueness of Lemma 25.1, $\sigma = \mathrm{id}$ and $x_i = z$ for all $i$. Therefore

$$\nu(z) = z^{|T|} = z^{(G:P)}.$$

Since $P$ is abelian, $[G,G] \subseteq \ker \nu$. Thus $\nu(z) = 1$, a contradiction, since $1 \neq z \in P$ and $z^{(G:P)} = 1$ implies that $z$ has order not divisible by $p$. □

Another application:

25.3. Proposition. *If $G$ is a group such that $Z(G)$ has finite index $n$, then $(gh)^n = g^n h^n$ for all $g, h \in G$.*

PROOF. Note that we may assume that $\alpha = \mathrm{id}$, as $Z(G)$ is abelian. Let $g \in G$. By Lemma 24.4 there are positive integers $n_1, \dots, n_k$ such that $n_1 + \cdots + n_k = n$ and elements $t_1, \dots, t_k$ of a transversal of $Z(G)$ in $G$ such that

$$\nu(g) = \prod_{i=1}^{k} t_i g^{n_1} t_i^{-1}.$$

Since $g^{n_i} \in Z(G)$ for all $i \in \{1, \dots, k\}$ (as $t_i g^{n_i} t_i^{-1} \in Z(G)$), it follows that

$$\nu(g) = g^{n_1 + \cdots + n_k} = g^n.$$

Now Theorem 24.2 implies the claim. □

The same idea implies the following property:

25.4. EXERCISE. If $G$ is a group and $K$ is a central subgroup of finite index $n$, then $(gh)^n = g^n h^n$ for all $g, h \in G$.

25.5. PROPOSITION. *Let $G$ be a finite group and $H$ a central subgroup of index $n$, where $n$ is coprime with $|H|$. Then $G \simeq N \rtimes H$.*

PROOF. Since $H$ is abelian, $H = H/[H, H]$. Let $\nu\colon G \to H$ be the transfer map and $h \in H$. By Lemma 24.4,

$$\nu(h) = \prod_{i=1}^{m} s_i^{-1} h^{n_i} s_i,$$

where each $s_i^{-1} h^{n_i} s_i \in H$. Since $h^{n_i} \in H \subseteq Z(G)$ for all $i$, it follows that $s_i^{-1} h^{n_i} s_i = h^{n_i}$ for all $i$. Thus

$$\nu(h) = \prod_{i=1}^{m} s_i^{-1} h^{n_i} s_i = \prod_{i=1}^{m} h^{n_i} = h^{\sum_{i=1}^{m} n_i} = h^n.$$

The composition $f\colon H \hookrightarrow G \xrightarrow{\nu} H$ is a group homomorphism. We claim that it is an isomorphism. It is injective: If $h^n = 1$, then $|h|$ divides both $|H|$ and $n$. Since $n$ and $|H|$ are coprime, $h = 1$. It is surjectice: Since $n$ and $|H|$ are coprime, there exists $m \in \mathbb{Z}$ such that $nm \equiv 1 \bmod |H|$. If $h \in H$, then $h^m \in H$ and $\nu(h^m) = h^{nm} = h$.

Let $N = \ker f$. We claim that $G = N \rtimes H$. By definition, $N$ is normal in $G$ and $N \cap H = \{1\}$. To show that $G = NH$ note that $|NH| = |N||H|$ and $G/N \simeq H$. □

25.6. EXERCISE. Let $H$ be a central subgroup of a finite group $G$. If $|H|$ and $|G/H|$ are coprime, then $G \simeq H \times G/H$.

We now present a nice application to infinite groups taken from Serre's book [14, 7.12].

25.7. THEOREM. *Let $G$ be a torsion-free group that contains a finite-index subgroup isomorphic to $\mathbb{Z}$. Then $G \simeq \mathbb{Z}$.*

PROOF. We may assume that $G$ contains a finite-index normal subgroup isomorphic to $\mathbb{Z}$. Indeed, if $H$ is a finite-index subgroup of $G$ such that $H \simeq \mathbb{Z}$, then $K = \cap_{x \in G} x H x^{-1}$ is a non-trivial normal subgroup of $G$ (because $K = \mathrm{Core}_G(H)$ and $G$ has no torsion) and hence $K \simeq \mathbb{Z}$ (because $K \subseteq H$) and $(G : K) = (G : H)(H : K)$ is finite. The action of $G$ on $K$ by conjugation induces a group homomorphism $\epsilon\colon G \to \mathrm{Aut}(K)$. Since $\mathrm{Aut}(K) \simeq \mathrm{Aut}(\mathbb{Z}) = \{-1, 1\}$, there are two cases to consider.

Assume first that $\epsilon = \mathrm{id}$. Since $K \subseteq Z(G)$, let $\nu\colon G \to K$ be the transfer homomorphism. By Proposition 25.3 (more precisely, by Exercise 25.4), $\nu(g) = g^n$, where $n = (G : K)$. Since $G$ has no torsion, $\nu$ is injective. Thus $G \simeq \mathbb{Z}$ because it is isomorphic to a subgroup of $K$.

Assume now that $\epsilon \neq \mathrm{id}$. Let $N = \ker \epsilon \neq G$. Since $K \simeq \mathbb{Z}$ is abelian, $K \subseteq N$. The result proved in the previous paragraph applied to $\epsilon|_N = 1$ implies that $N \simeq \mathbb{Z}$, as $N$ contains a finite-index subgroup isomorphic to $\mathbb{Z}$. Let $g \in G \setminus N$. Since $N$ is normal in $G$, $G$ acts by conjugation on $N$ and hence there exists a group homomorphism $c_g \in \mathrm{Aut}(N) \simeq \{-1, 1\}$. Since $K \subseteq N$ and $g$ acts non-trivially on $K$,

$$c_g(n) = gng^{-1} = n^{-1}$$

for all $n \in N$. Since $g^2 \in N$,

$$g^2 = gg^2g^{-1} = g^{-2}.$$

Therefore $g^4 = 1$, a contradiction since $g \neq 1$ and $G$ has no torsion. □

## 26. Dietzman's theorem

26.1. Theorem (Dietzmann). *Let $G$ be a group and $X \subseteq G$ be a finite subset of $G$ closed by conjugation. If there exists $n$ such that $x^n = 1$ for all $x \in X$, then $\langle X \rangle$ is a finite subgroup of $G$.*

Proof. Let $S = \langle X \rangle$. Since $x^{-1} = x^{n-1}$, every element of $S$ can be written as a finite product of elements of $X$. Fix $x \in X$. We claim that if $x \in X$ appears $k \geq 1$ times in the word $s$, then we can write $s$ as a product of $m$ elements of $X$, where the first $k$ elements are equal to $x$. Suppose that

$$s = x_1x_2 \cdots x_{t-1}xx_{t+1} \cdots x_m,$$

where $x_j \neq x$ for all $j \in \{1, \ldots, t-1\}$. Then

$$s = x(x^{-1}x_1x)(x^{-1}x_2x) \cdots (x^{-1}x_{t-1}x)x_{t+1} \cdots x_m$$

is a product of $m$ elements of $X$ since $X$ is closed under conjugation and the first element is $x$. The same argument implies that $s$ can be written as

$$s = x^k y_{k+1} \cdots y_m,$$

where each $y_j$ belongs to $X \setminus \{x\}$.

Let $s \in S$ and write $s$ as a product of $m$ elements of $X$, where $m$ is minimal. We need to show that $m \leq (n-1)|X|$. If $m > (n-1)|X|$, then at least one $x \in X$ appears exactly $n$ times in the representation of $s$. Without loss of generality, we write

$$s = x^n x_{n+1} \cdots x_m = x_{n+1} \cdots x_m,$$

a contradiction to the minimality of $m$. □

## 27. Schur's commutator theorem

27.1. Theorem (Schur). *Let $G$ be a group. If $Z(G)$ has finite index in $G$, then $[G, G]$ is finite.*

Proof. Let $n = (G : Z(G))$ and $X$ be the set of commutators of $G$. We claim that $X$ is finite, in fact $|X| \leq n^2$. A routine calculation shows that the map

$$\varphi\colon X \to G/Z(G) \times G/Z(G), \quad [x, y] \mapsto (xZ(G), yZ(G)),$$

is well-defined. It is, moreover, injective: if $(xZ(G), yZ(G)) = (uZ(G), vZ(G))$, then $u^{-1}x \in Z(G)$, $v^{-1}y \in Z(G)$. Thus

$$[u,v] = uvu^{-1}v^{-1} = uv(u^{-1}x)x^{-1}v^{-1} = xvx^{-1}(v^{-1}y)y^{-1} = xyx^{-1}y^{-1} = [x,y].$$

Moreover, $X$ is closed under conjugation, as

$$g[x,y]g^{-1} = [gxg^{-1}, gyg^{-1}]$$

for all $g, x, y \in G$. Since $G \to Z(G)$, $g \mapsto g^n$ is a group homomorphism, Proposition 25.3 implies that $[x,y]^n = [x^n, y^n] = 1$ for all $[x,y] \in X$. The theorem follows from applying Dietzmann's theorem. □

27.2. Exercise. Let $G$ be the group with generators $a, b, c$ and relations $ab = ca$, $ac = ba$ and $bc = ab$. Prove the following statements:

**1)** $G$ is infinite and non-abelian.
**2)** $Z(G)$ has finite index in $G$ and every conjugacy class of $G$ is finite.
**3)** $[G,G]$ is finite.
**4)** The subgroup $N = \langle a^2 \rangle$ of $G$ generated by $a^2$ is central and $G/N$ is finite.

We conclude the section with some results similar to that of Schur.

27.3. Theorem (Niroomand). *If the set of commutators of a group $G$ is finite, then $[G,G]$ is finite.*

Proof. Let $C = \{[x_1, y_1], \ldots, [x_k, y_k]\}$ be the (finite) set of commutators of $G$ and

$$H = \langle x_1, x_2, \ldots, x_k, y_1, y_2, \ldots, y_k \rangle.$$

Since $C$ is a set of commutators of $H$, it follows that $[G,G] = \langle C \rangle \subseteq [H,H]$. To simplify the notation we write $H = \langle h_1, \ldots, h_{2k} \rangle$. Since $h \in Z(H)$ if and only if $h \in C_H(h_i)$ for all $i \in \{1, \ldots, 2k\}$, we conclude that $Z(H) = C_H(h_1) \cap \cdots \cap C_H(h_{2k})$. Moreover, if $h \in H$, then $hh_ih^{-1} = ch_i$ for some $c \in C$. Thus the conjugacy class of each $h_i$ contains at most as many elements as $C$. This implies that

$$|H/Z(H)| = |H/\cap_{i=1}^{2k} C_H(h_i)| \leq \prod_{i=1}^{2k} (H : C_H(h_i)) \leq |C|^{2k}.$$

Since $H/Z(H)$ is finite, $[H,H]$ is finite. Hence $[G,G] = \langle C \rangle \subseteq [H,H]$ is a finite group. □

27.4. Theorem (Hilton–Niroomand). *Let $G$ be a finitely generated group. If $[G,G]$ is finite and $G/Z(G)$ is generated by $n$ elements, then*

$$|G/Z(G)| \leq |[G,G]|^n.$$

Proof. Assume that $G/Z(G) = \langle x_1 Z(G), \ldots, x_n Z(G) \rangle$. Let

$$f \colon G/Z(G) \to [G,G] \times \cdots \times [G,G], \quad y \mapsto ([x_1, y], \ldots, [x_n, y]).$$

Note that $f$ is well-defined: If $y \in G$ and $z \in Z(G)$, then $[x_i, y] = [x_i, yz]$ for all $i$. Then $f(yz) = f(y)$.

The map $f$ is injective. Assume that $f(xZ(G)) = f(yZ(G))$. Then $[x_i, x] = [x_i, y]$ for all $i \in \{1, \ldots, n\}$. For each $i$ we compute

$$\begin{aligned}[x^{-1}y, x_i] &= x^{-1}[y, x_i]x[x^{-1}, x_i] \\ &= x^{-1}[y, x_i][x_i, x]x = x^{-1}[x_i, y]^{-1}[x_i, x]x = x^{-1}[x_i, y]^{-1}[x_i, y]x = 1.\end{aligned}$$

This implies that $x^{-1}y \in Z(G)$. Indeed, since every $g \in G$ can be written as $g = x_k z$ for some $k \in \{1, \dots, n\}$ and some $z \in Z(G)$, it follows that

$$[x^{-1}y, g] = [x^{-1}y, x_k z] = [x^{-1}y, x_k] = 1.$$

Since $f$ is injective, $|G/Z(G)| \leq |[G, G]|^n$. □

27.5. Exercise. Prove Theorem 27.4 from Theorem 27.3.

## 28. Burnside normal complement theorem

We first need a way of computing the transfer map.

28.1. Lemma. *Let $G$ be a group and let $H$ be a subgroup of index $n$. Let $T = \{t_1, \dots, t_n\}$ be a transversal of $H$ in $G$. For each $g \in G$, there exists $m \geq 1$ and there exist $s_1, \dots, s_m \in T$ and positive integers $n_1, \dots, n_m$ such that $s_i^{-1} g^{n_i} s_i \in H$, $n_1 + \cdots + n_m = n$, and*

$$\nu(g) = \prod_{i=1}^{m} s_i^{-1} g^{n_i} s_i.$$

Proof. For each $i$, there exist $h_1, \dots, h_n \in H$ and $\sigma \in \mathbb{S}_n$ such that $gt_i = t_{\sigma(i)} h_i$. We write $\sigma$ as a product of disjoint cycles

$$\sigma = \alpha_1 \cdots \alpha_m.$$

Fix $i \in \{1, \dots, n\}$ and write $\alpha_i = (j_1 \cdots j_{n_i})$. Since

$$gt_{j_k} = t_{\sigma(j_k)} h_{j_k} = \begin{cases} t_{j_1} h_{n_k} & \text{if } k = n_i, \\ t_{j_{k+1}} h_k & \text{otherwise,} \end{cases}$$

then

$$\begin{aligned} t_{j_1}^{-1} g^{n_i} t_{j_1} &= t_{j_1}^{-1} g^{n_i - 1} g t_{j_1} \\ &= t_{j_1}^{-1} g^{n_i - 1} t_{j_2} h_1 \\ &= t_{j_1}^{-1} g^{n_i - 2} g t_{j_2} h_1 \\ &= t_{j_1}^{-1} g^{n_i - 2} t_{j_3} h_2 h_1 \\ &\vdots \\ &= t_{j_1}^{-1} g t_{j_{n_i}} h_{n_{i-1}} \cdots h_2 h_1 \\ &= t_{j_1}^{-1} t_{j_1} h_{n_i} \cdots h_2 h_1 \in H. \end{aligned}$$

We then take $s_i = t_{j_1} \in T$. The result is thus demonstrated by observing that $\nu(g) = h_1 \cdots h_n$. □

28.2. Lemma. *Let $G$ be a finite group and let $p$ be a prime dividing the order of $G$. Let $P \in \mathrm{Syl}_p(G)$. If $g, h \in C_G(P)$ are conjugate in $G$, then they are conjugate in $N_G(P)$.*

Proof. Let $x \in G$ such that $g = xhx^{-1}$. Then $g \in C_G(xPx^{-1})$. Hence $P$ and $xPx^{-1}$ are Sylow subgroups of $C_G(g)$. By Sylow's theorem, there exists $c \in C_G(g)$ such that $P = cxP(cx)^{-1}$. Then $cx \in N_G(P)$ and

$$(cx)h(cx)^{-1} = c(xhx^{-1})c^{-1} = cgc^{-1} = g.$$ □

28.3. Definition. Let $G$ be a finite group and let $p$ be a prime dividing the order of $G$. A **$p$-normal complement** is a normal subgroup $N$ of $G$ of order coprime to $p$ such that $(G : N)$ is a power of $p$.

28.4. Definition. A finite group is said to be **$p$-nilpotent** if it has a $p$-normal complement.

28.5. Proposition. *Let $G$ be a finite group with a $p$-normal complement $N$. Then $N$ is a characteristic subgroup of $G$.*

PROOF. Suppose $|G| = p^\alpha n$, where $n$ is coprime to $p$, and let $\pi\colon G \to G/N$ be the canonical homomorphism. By hypothesis, $N$ has order $n$. We will show that $N$ is the unique subgroup of $G$ of order $n$. If $K$ is a subgroup of $G$ of order $n$, then $\pi(K) \simeq K/K \cap N$ and hence the order of $\pi(K)$ divides $n$. But also the order of $\pi(K)$ divides the prime $p$ since $\pi(K) \leq G/N$. Thus $\pi(K)$ is trivial and so $K = N$, implying that $G$ has a unique subgroup of order $n$. In particular, $N$ is a characteristic subgroup of $G$. □

28.6. THEOREM (Burnside's normal complement theorem). *Let $G$ be a finite group and let $p$ be a prime dividing $|G|$. Let $P \in \mathrm{Syl}_p(G)$ be such that $P \subseteq Z(N_G(P))$. Then $G$ is $p$-nilpotent.*

PROOF. The group $P$ is abelian. Let $\nu\colon G \to P$ be the transfer homomorphism and $g \in P$. By Lemma 28.1, there exist $s_1, \ldots, s_m \in G$ and there exist $n_1, \ldots, n_m$ such that $n_1 + \cdots + n_m = n$, $s_i^{-1} g^{n_i} s_i \in P$, and

$$v(g) = \prod_{i=1}^{m} s_i^{-1} g^{n_i} s_i.$$

Since $P$ is abelian, $P \subseteq C_G(P)$. By Lemma 28.2, there exist elements $c_i \in N_G(P)$ such that

$$s_i^{-1} g^{n_i} s_i = c_i^{-1} g^{n_i} c_i.$$

Then $s_i^{-1} g^{n_i} s_i = g_i^{n_i}$, since $P \subseteq Z(N_G(P))$. Thus $\nu(g) = g^n$, where $n = (G : P)$. Since $n$ and $|P|$ are coprime, there exist $r, s \in \mathbb{Z}$ such that $rn + s|P| = 1$. This implies that the restriction $\nu|_P$ is surjective since

$$g = (g^r)^n = \nu(g^r).$$

By the isomorphism theorem,

$$P/\ker \nu \cap P \simeq \nu(P) = P.$$

Thus $\ker \nu \cap P = \{1\}$. Moreover, $\nu(G) = \nu(P)$, since $P \supseteq \nu(G) \supseteq \nu(P) = P$.

We will show that $\ker \nu$ is a $p$-normal complement in $G$. Clearly, $\ker \nu$ is normal in $G$. Since $(G : \ker \nu) = |\nu(G)| = |P|$ and $P$ is a Sylow $p$-subgroup, we conclude that $\ker \nu$ has order coprime to $p$. □

28.7. EXERCISE. Let $G$ be a group and $H$ a subgroup of $G$. Prove that $C_G(H)$ is a normal subgroup of $N_G(H)$ and $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\mathrm{Aut}(H)$.

28.8. EXERCISE. Let $G$ be a finite group and let $p$ be the smallest prime dividing $|G|$. If some $P \in \mathrm{Syl}_p(G)$ is cyclic, then $G$ is $p$-nilpotent.

28.9. EXERCISE. Let $G$ be a finite group such that all its Sylow subgroups are cyclic. Then $G$ is solvable.

Let us prove something stronger:

28.10. PROPOSITION. *Let $G$ be a finite group such that all its Sylow subgroups are cyclic. Then $G$ is super-solvable.*

PROOF. Suppose $G$ is nontrivial and Let us induct on $|G|$. If $p$ is the smallest prime dividing $|G|$, by Corollary 28.8 the group $G$ has a normal $p$-complement $N$. By the inductive hypothesis, $N$ is solvable. Since $G/N$ is a $p$-group, it is solvable. Hence $G$ is solvable. □

28.11. Exercise. Let $G$ be a finite group whose order is square-free. Prove that $G$ is solvable.

28.12. Corollary. *Let $G$ be a non-abelian finite simple group and let $p$ be the smallest prime dividing $|G|$. Then either $p^3$ divides $|G|$ or 12 divides $|G|$.*

Proof. Let $P \in \mathrm{Syl}_p(G)$. By Corollary 28.8, $P$ is not cyclic, so $|P| \geq p^2$. If $p^3$ does not divide $|G|$, then $P \simeq C_p \times C_p$ is a $\mathbb{F}_p$-vector space of dimension two. Since $|N_G(P)/C_G(P)|$ divides the order of $G$, the prime divisors of $|N_G(P)/C_G(P)|$ are at least $p$. Moreover, as $N_G(P)/C_G(P)$ is isomorphic to a subgroup of $\mathrm{Aut}(P)$ by Exercise 28.7, and $\mathrm{Aut}(P) \simeq \mathbf{GL}_2(p)$ has order

$$(p^2-1)(p^2-p) = p(p+1)(p-1)^2,$$

it follows that $|N_G(P)/C_G(P)|$ divides $p(p+1)(p-1)^2$. Since $P$ is abelian, $P \subseteq C_G(P)$. Thus $|N_G(P)/C_G(P)|$ is coprime to $p$, so $|N_G(P)/C_G(P)|$ divides $(p+1)(p-1)^2$. Since $p$ is the smallest prime dividing $|G|$, $p-1$ and $|N_G(P)/C_G(P)|$ are coprime, implying that $|N_G(P)/C_G(P)|$ divides $p+1$. Furthermore, by Theorem 28.6, $|N_G(P)/C_G(P)| \neq 1$. This implies that $p = 2$, because if $p$ is odd, the smallest prime dividing $|N_G(P)/C_G(P)|$ is at least $p+2$. Thus, $p = 2$, and consequently $|N_G(P)/C_G(P)| = 3$. Hence $|G|$ is divisible by $12 = 2^2 3$. □

28.13. Theorem. *Let $G$ be a finite group and let $P$ be an abelian Sylow subgroup. Then $[G,G] \cap P \cap Z(N_G(P)) = \{1\}$.*

Proof. Let $x \in [G,G] \cap P \cap Z(N_G(P))$ and let $\nu\colon G \to P$ be the transfer homomorphism. By Lemma 28.1, there exist $s_1, \ldots, s_m \in G$ and $n_1, \ldots, n_m$ such that $n_1 + \cdots + n_m = (G:P)$, $s_i^{-1} g^{n_i} s_i \in P$, and

$$v(x) = \prod_{i=1}^{m} s_i^{-1} x^{n_i} s_i.$$

Since $P$ is abelian, $P \subseteq C_G(P)$. Then $x^{n_i}$ and $s_i^{-1} x^{n_i} s_i$ are conjugate in $N_G(P)$ by Lemma 28.2. Since $x^{n_i}$ is central in $N_G(P)$ and $[G,G] \subseteq \ker \nu$, it follows that $x = 1$ since $1 = \nu(x) = x^{(G:P)}$ and $x \in P$. □

28.14. Corollary. *Let $G$ be a non-abelian finite group and let $P \in \mathrm{Syl}_2(G)$ such that*

$$P \simeq C_{a_1} \times \cdots \times C_{a_k}$$

*with $a_1 > a_2 \geq a_3 \geq \cdots \geq a_k \geq 2$. Then $G$ is not simple.*

Proof. Let $S = \{x^{n/2} : x \in P\}$. It's easy to see that $S$ is a subgroup of $P$ and $S$ is characteristic in $P$, i.e., $f(S) \subseteq S$ for every $f \in \mathrm{Aut}(P)$. Since $S \simeq C_2$, we can write $S = \{1, s\}$. Then $s \in Z(N_G(P))$ because $gsg^{-1} \in S$ for every $g \in N_G(P)$. By Theorem 28.13, $s \notin [G,G]$, so $[G,G] \neq G$. If $G$ were simple, then $G$ would be abelian since $[G,G] = 1$. □

Finite groups whose Sylow subgroups are cyclic are solvable.

28.15. Definition. A **Z-group** is a finite group $G$ such that all its Sylow subgroups are cyclic.

28.16. Definition. A group $G$ is called **meta-cyclic** if $G$ has a normal cyclic subgroup $N$ such that $G/N$ is cyclic.

28.17. Exercise. If $G$ is a solvable group, then $C_G(F(G)) = F(G)$.

28.18. Theorem. *Every Z-group is meta-cyclic.*

Proof. Let $G$ be a Z-group. By Proposition 28.10, $G$ is solvable and hence the Fitting subgroup $F(G)$ satisfies $C_G(F(G)) \subseteq F(G)$.

Let us show that $F(G)$ is cyclic. Since $F(G)$ is nilpotent, $F(G)$ is the direct product of its Sylow subgroups. Since every Sylow subgroup of $F(G)$ is a $p$-subgroup of $G$, every Sylow subgroup of $F(G)$ is cyclic (as it is contained in some Sylow subgroup of $G$).

Since $F(G)$ is cyclic, $F(G)$ is abelian and hence $F(G) \subseteq C_G(F(G))$. If $G$ acts on $F(G)$ by conjugation, then there is a homomorphism $\gamma\colon G \to \mathrm{Aut}(F(G))$ such that $\ker\gamma = C_G(F(G)) = F(G)$ (since $\gamma_g(x) = gxg^{-1}$). In particular, $G/F(G)$ is abelian as it is isomorphic to a subgroup of the abelian group $\mathrm{Aut}(F(G))$. Moreover, since the Sylow subgroups of $G/F(G)$ are cyclic (as they are quotients of Sylow subgroups of $G$), $G/F(G)$ is cyclic. □

## 29. Miller's double cosets theorem

Let $G$ be a group and $H$ and $K$ be subgroups of $G$. The group $L = H \times K$ acts on $G$ by

$$(h,k)\cdot g = hgk^{-1}, \quad h \in H, k \in K, g \in G.$$

The orbits of this action are the set of the form

$$HgK = \{hgk : h \in H,\, k \in K\}.$$

A set of the form $HgK$ for some $g \in G$ is called a **double coset** modulo $(H,K)$ with representative $g$. In particular, any two double cosets are either disjoint or equal, and $G$ decomposes as a disjoint union

$$G = \bigcup_{i\in I} Hg_iK,$$

for some set $I$. Let

$$L_g = \{(h,k) \in H \times K : hgk^{-1} = g\} = \{(h, g^{-1}hg) \in H \times K\}.$$

Then $|L_g| = |H \cap gKg^{-1}|$, because there is a bijection $L_g \to H \cap gKg^{-1}$. By the fundamental counting principle,

$$|HgK| = (L : L_g) = \frac{|H \times K|}{|H \cap gKg^{-1}|} = \frac{|H||K|}{|H \cap gKg^{-1}|}.$$

We need a lemma.

29.1. Lemma. *Let $G$ be a finite group, $x \in G$, and $H$ and $K$ be subgroups of $G$. Then*

$$\#\{zK : zK \subseteq HxK\} = (H : xKx^{-1} \cap H).$$

Proof. Let $L = xKx^{-1} \cap H$ and

$$\varphi\colon H/L \to \{yK : yK \subseteq HxK\}, \quad hL \mapsto hxK.$$

The map $\varphi$ is well-defined. If $hL = h_1L$, then $h^{-1}h_1 \in L$. Thus $h^{-1}h_1 = xkx^{-1}$ for some $k \in K$. This means that

$$(h_1x)^{-1}(hx) = x^{-1}h_1^{-1}hx = k^{-1} \in K,$$

that is $\varphi(hL) = hxK = h_1xK = \varphi(h_1L)$.

The map $\varphi$ is surjective: If $zK$ is such that $zK \subseteq HxK$, then $z = hxk$ for some $k \in K$. In particular, $zK = hxK$. Now $\varphi(hL) = hxK = zK$.

The map $\varphi$ is injective: If $hxK = h_1xK$, then $x^{-1}h_1^{-1}hx \in K$. Moreover, $h_1^{-1}h \in xKx^{-1} \cap H = L$. Thus $h_1L = hL$. □

29.2. Exercise. Let $G$ be a finite group, $H$ and $K$ be subgroups of $G$, and $x \in G$. Prove that

$$\#\{Hy : Hy \subseteq HxK\} = (K : xHx^{-1} \cap K).$$

29.3. Theorem (Miller). *Let $G$ be a finite group and $H$ and $K$ be subgroups of $G$ of the same index. Then there exists a common complete set of representatives for the right cosets of $H$ in $G$ and the left cosets of $K$ in $G$.*

PROOF. Let $Hy$ be a right coset and $zK$ be a left coset. Note that $Hy$ and $zK$ have a common representative if and only if $Hy \cap zK \neq \emptyset$, as

$$Hy = Hx \text{ and } zK = xK \iff xy^{-1} \in H \text{ and } z^{-1}x \in K \iff x \in Hy \cap zK.$$

The group $G$ decomposes as a disjoint union of finitely many double cosets. Each double coset $HxK$ is a disjoint union of finitely many right cosets of $H$ and a disjoint union of finitely many left cosets of $K$. Thus

$$HxK = \bigcup_{i=1}^{k} Hy_i = \bigcup_{j=1}^{l} z_j K,$$

where the unions are disjoint. Since $|H| = |K|$, by applying cardinality, it follows that $k = l$. To prove the theorem it is enough to show that each $Hy_i$ intersects every $z_j K$.

Note that for each $i \in \{1, \ldots, k\}$ there exists $j \in \{1, \ldots, k\}$ such that $Hy_i \cap z_j K \neq \emptyset$. Without loss of generality, we may assume (reordering if needed) that $Hy_1 \cap z_j K \neq \emptyset$ for all $j \in \{1, \ldots, m\}$, where $1 \leq m \leq k$. Then

$$Hy_1 \subseteq \bigcup_{j=1}^{m} z_j K.$$

Then

$$Hy_1 K \subseteq \bigcup_{j=1}^{m} z_j K \subseteq \bigcup_{j=1}^{k} z_j K = HxK.$$

Since $Hy_1 K$ and $HxK$ are double cosets with non-empty intersection, they are equal. Thus

$$|HxK| = |Hy_1 K| \leq \sum_{j=1}^{m} |z_j K| = m|K|.$$

By Lemma 29.1,

$$k = \#\{z_j K : z_j K \subseteq HxK\} = (H : xKx^{-1} \cap H).$$

Therefore

$$k|K| = \frac{|H||K|}{|H \cap xKx^{-1}|} = |HxK| \leq m|K|$$

and hence $k = m$. □

29.4. EXERCISE (Hall). Let $G$ be a finite group and $H$ be a subgroup of $G$ with $(G : H) = n$. Then there exists $x_1, \ldots, x_n \in G$ such that $\{Hx_1, Hx_2, \ldots, Hx_n\} = \{x_1 H, x_2 H, \ldots, x_n H\}$.

## Some topics for final projects

We collect here some topics for final presentations. Some topics can also be used as bachelor's or master's theses.

**Formanek's theorem.** Kaplansky zero divisors conjecture (Conjecture 14.1) holds for super-solvable groups. This is Formanek's theorem, see Theorem 14.2.

**Carter subgroups.** The existence of Carter subgroups appears in Theorem **??**. For the proof of Carter's result, we refer to [3].

**Itô's theorem.** This nice theorem is surprisingly easy to prove. See Theorem **??**. There is a generalization found by Sysak, see Theorem **??**.

**The Mann subgroup.** How the set of sizes of conjugacy classes of a finite group influence the structure of the group? A partial answer goes back to Mann. See Theorem **??**.

**The Chermak–Delgado subgroup.** In [4], Chermak and Delgado presented a measuring argument for finite groups. They obtained some results about special families of finite groups, and powerful applications to finite simple groups and finite p-groups. See Theorem **??**.

**Hall's subgroups.** For finite solvable groups, Hall developed a theory similar to that of Sylow. The proofs presented in Theorems 16.3 and 16.6 (see page 38) use the Schur–Zassenhaus theorems.

**Hall's marriage theorem.** Given a set of $n$ employees, fill out a list of the jobs each of them would be able to preform. We can give each person the "perfect" job if and only if for every $k \in \{1, \dots, n\}$ the union of any $k$ of the lists contains at least $k$ jobs.

For a proof of the Hall marriage theorem, see [7]. The theorem is equivalent to several other powerful theorems in combinatorics. Hall's theorem provides a combinatorial way of solving Exercise 29.4.

**The Wielandt automorphism tower theorem.** One striking application of subnormality is a beautiful result of Wielandt about the automorphism tower of a finite group with trivial center. See Theorem **??**.

**Gardam's theorem.** Let $K$ be a field and $G$ be a torsion-free group. What do the units of $K[G]$ look like? Gardam solved the Kaplansky units problem, answering negatively Question **??**. See Theorem **??** for his solution for fields of characteristic two.

**Passman's theorem.** There are several interesting open problems in the theory of group algebras. One of these problems is about zero-divisors in group algebras and another one is about group algebras being reduced as rings. Passman's theorem states that these problems are equivalent. See Theorem **??**.

**The Alperin–Kuo theorem.** One way to prove the Alperin–Kuo theorem (see Theorem **??**) is to combine ring-theoretical tools with the transfer homomorphism. The original proof uses group cohomology; see [1].

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